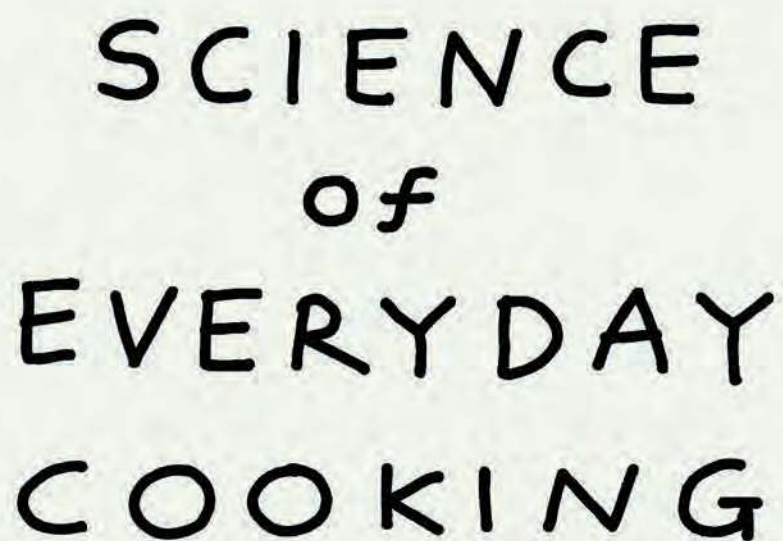




SCIENCE
of
EVERYDAY
COOKING

Science of Everyday Cooking

Recipes and Reflections from Cooking Experiments



Final projects by students of YSC 1223

December 2022

YaleNUSCollege

Authors:

Aaron Sean Tan Zhenjie

Alvin Shrestha

Amanda Bianca Chen

Anthea Cheo

Arjav Devangkumar Sanghavi

Barnabas Mah

Brian Tan Yi-Shue

Carine Chan Sze Ching

Chan Yi Qian

Chua Yi Jia

Dylan Lee Pak Han

Edmund Chow Wei Jie

Gabrielle Charis Alano Tagtag

Gunnar Hoanh Son Schmidt

Harindrar S/o Jayanthinathan

Howe Ying Yi

John Ng Cheng Earn

Kristen Lauren Oliveira

Lim Yan Zhen

Lucas Puah Jia Rong

Martin Choo Yi Kang

Max Dilan Lopez Toledano

Mohammad Dabeer Ahmed

Mohammad Zirdi Syukur

Ng Kye Been

Nur Aina Binte Sapari

Ozair Faisal

Rachel Cheong

Raphael Antonio Liboro Romasanta

Sewen Thy

Silky Siqi Hou

Tsega (Pache) Yilma Abebe

Woo Hoi Yuet

Yelani Sawithra Bopitiya

Yu Chi Ying Meryl

Zen Alexander Goh Wei Han

Front and back cover designer:

Chan Yi Qian, Ding Manqi Angela

Editor:

Zhuang Bilin

*To the Yale-NUS Community,
which is full of passion and dedication*

Contents



Choco Cube	1
Panna Cotta	9
Cream Puffs	19
Matcha Basque Cheesecake	33
Mochi Donuts	42
Apple Crumble	48
Aquafaba Meringues	66
Pesto Sauce	70
Mala Xiang Guo	76
Marinating Beef Steak	86
Butter Chicken	93
Hand-pulled Noodles	98
The Perfect Eggplant	104

Preface



Cooking is such a delicious way to teach science. For me, teaching YSC1123 *Science of Everyday Cooking* at Yale-NUS College is a dream come true. This course is a perfect combination of my passions. I had the opportunity to use cooking to teach science, which was so much fun. At the same time, I have learned lots of science by teaching this course, which (hopefully) made my cooking better. In one semester, we had much fun doing weekly cooking experiments in the Saga buttry. We made mayonnaise and looked at it under the microscope, tempered chocolate in different ways to get the tastiest phase transition, and made ice cream by tossing the ultra-cold salt-ice mixture in the buttry. We also traveled to Brettschneider's Baking & Cooking School to learn the art and science of bread from the best chefs. This course has been a memorable journey for me. I hope it has been a memorable experience for the students in this course, too.

This book is an effort to keep the memories alive. It is a compilation of cooking tips and tricks discovered by the students in their final project. Each team came up with a culinary question they wanted to tackle and then did a cooking experiment to answer it. This book documented what they found. It included apple crumble optimized by Fourier transform, panna cotta with the best elasticity, choco cube with the perfect phase transition (in short, melts) at the center, and many more. I hope you enjoy the discoveries in this book!

Zhuang Bilin

February 2023 in Singapore

Choco Cube



By Aina, Lucas, & Martin



Preparation time 3.5 hours, including resting time



Baking time 30 minutes



Serves 6–8

Summary

Ever wished that you could dip your hot fresh bread in rich decadent chocolate, but you have no time to make a dipping sauce? Well, fret not — with this inventive recipe, the bread will already come with chocolate! The *Choco Cube* has that crackling exterior yet fluffy interior we all know and love but with a wonderful surprise — a soft, melted center filled with molten chocolate! Who said you can't have dessert for breakfast?

With a basic bread recipe, we experimented with ingredients and proportions to find out:

1. How do we make bread fluffier?
2. How do we ensure the chocolate inside the bread cube melts?

We found that fluffy bread would require a higher fat content to be incorporated into the dough. Additionally, to get rich chocolate that flows out of the bread is to use a standard ganache mixture, a 1:1 ratio of chocolate and cream which can freeze and melt easily.

Why this project

We were very inspired by the breadmaking workshop we did at Brettschneider's Baking & Cooking School. It is incredible to think that you only need two basic ingredients — flour and water — to make over 200 different types of bread! So with a limited budget and a desire to test our bread-making skills on our own, we scoured the Internet, and our minds, for a simple culinary product that would be a spectacle to behold — hence, the birth of *Choco Cube*.

The bread we made for this project is interesting as it relies on conventional gluten network formation but incorporates fat into this process to restrict the formation of long gluten strands, allowing for softer and fluffier bread. Additionally, this recipe takes into account scientific processes like heat transfer in ensuring the chocolate in the middle of the bread is able to remain molten and gooey. Finding the right chocolate ganache mixture to use ensures that the chocolate remains rich and thick, without separating into fat and chocolate solids due to the high heat in the oven.

Our Experiment

Our bread recipe is based on Japanese milk bread called *shokupan*. Instead of regular all-purpose flour, we used bread flour to ensure higher protein content to create more gluten networks. We also used a cube bread mould instead of a regular bread loaf tin. Lastly, unlike previous recipes which used oil, we added butter and milk as fats substitute for this recipe.

For the ganache, we specifically bought couverture chocolate chips rather than regular chocolate bars to ensure there were no added ingredients or preservatives that would interfere with the melting, freezing, and baking processes. Using the same bread recipe, we varied the ratio of chocolate to cream in the ganache as such:

1. Pure chocolate
2. 2 x chocolate : 1 x cream
3. 1 x chocolate : 1 x cream
4. 1 x chocolate : 2 x cream

Our Discovery

As mentioned earlier, bread flour has a higher percentage (12-14%) of proteins. We found that the higher amount of gluten in bread flour makes for a more elastic dough than the ones we kneaded previously, which produces lighter and chewier bread.

We also found that the addition of fats – from butter and milk – can make the bread more tender, moist, and fine-grained. The fat coats the gluten strands, which slows down the yeast fermentation process. It also lubricates the dough, allowing the gluten networks to expand easily. Fats are also what make the exterior of the bread soft and not as crispy as some other European breads.



The fluffier shokupan bread (left) with fewer defined crumbs compared to a fougasse (right).

We also observed different levels of melt when it came to the ganache after pulling open the bread cubes.



Pure chocolate (top left), 1:2 chocolate to cream (top right), 1:1 chocolate to cream (bottom left), 1:2 chocolate to cream (bottom right).

As seen in the pictures above, the top left and bottom right were the ‘worst’ versions of the Choco Cube. The former had chocolate that separated into coagulated cocoa solids and fats after baking. One possible reason why this first loaf had the weird chocolate chunks that are not present in other loaves was because it is not part of a stabilised mixture. Needless to say, it failed to provide the molten chocolate experience.

However, the 1:2 chocolate-to-cream ganache had the opposite problem. It was way too watery and made for a very difficult eating experience. Thus, even though it was part of a stable mixture, the ratios were off for this particular recipe.

Thus, the top right and bottom left loaves were arguably the better products. The chocolate-to-cream ratio was just right such that there was a satisfying ooze when the loaf was opened up but it was not too creamy that it created a mess. Depending on preference, either ratio would have been a good amount.

Conclusion

This is the amazing final product of our Choco Cube experiment:



So fluffy! So gooey!

Through this experiment, we found out the components of fluffy bread and the optimal levels of chocolate to cream in a ganache filling for bread. In terms of bread, it requires fats to add flavour, make the crust tender, and give it a soft exterior. Typically, we would add fats by including milk or butter, or both. This technique is quite common in Asian-type breads. Hence, the taste should be quite familiar to Singaporeans who love buying sweet bread from their local bakeries.

For the ganache, we played around with the ratios of chocolate to cream. We found that by combining the chocolate with cream, we are able to make a stable mixture that would not separate after melting, freezing, and baking. This is why the loaves, aside from the pure chocolate loaves, were able to provide the molten experience. The next step was to determine the ideal type of melt which we have concluded to be the 1:1 ratio of chocolate to cream.

In the future, it would be interesting to see how other types of fillings would work with this recipe! A delicious curry (with coconut cream) or a hearty seafood stew might work just as well! Particularly, we hope that this might be a revolutionary culinary invention of an accessible meal for schoolkids, working persons, and the elderly. It's easy to make, eat, and share – what else can't a perfect meal do?

Ingredients

Ganache, makes more than needed

✻ 300ml whipping cream ✻ 300g dark chocolate chips ✻ 50g unsalted butter

Shokupan

✻ 500g bread flour, plus more for dusting ✻ 24g white sugar ✻ 6g

salt ✨ 270ml water ✨ 100ml milk ✨ 6g instant yeast ✨ 50g unsalted butter, room temperature, cubed ✨ Nonstick cooking spray, for greasing

Special Equipment

✨ 4x4x4-inch cube bread mould

Directions

Ganache

1. Line a tray or bowl with parchment paper.
2. Put the chocolate and butter in a bowl.
3. Bring cream to a simmer over medium heat, just before boiling.
4. Pour cream over chocolate and butter and let sit for 2 minutes without stirring.
5. After 2 minutes, stir the mixture until well blended.
6. Pour onto the prepared tray or bowl. Once cooled to room temperature, place in the freezer for at least 2 hours.
7. Once frozen, cut into cubes and place back in the freezer.

Shokupan

1. Add the bread flour, sugar, and salt to a large bowl. Mix well.
2. Add the water, milk, and yeast to the flour mix. Mix with a bowl scraper until a dough forms.
3. Turn the dough onto a lightly floured surface and knead until smooth, for 5-10 minutes.

4. Flatten out the dough, then add the cubed butter to the center and wrap the dough around it. Continue kneading the dough until the butter is well-incorporated, about 10 minutes, or until the dough is smooth and shiny. Shape the dough into a ball, transfer it to a medium bowl, and cover it with plastic wrap. Let rise in a warm, draft-free place until doubled in size, about 1 hour.
5. Turn the proofed dough onto a lightly floured surface. Punch down the dough. Split the dough into 3-4 equal portions, to your liking. Roll each into a ball and cover with a damp towel and rest it for 20 minutes.
6. After 20 minutes, flatten out the dough with a rolling pin. Fill with 50-80g of frozen ganache cubes. Tuck the dough around the ganache, forming a ball, and pinch close the seam. Repeat for all dough balls.
7. Grease a 4x4x4-inch cube *shokupan* mold with nonstick spray. Place one dough ball for each mold. Cover with plastic wrap and proof in a warm, draft-free place for 1 hour, or until doubled in volume.
8. Preheat the oven to 425°F (220°C).
9. Close the lid of the mold and transfer it to the oven. Reduce the oven temperature to 400°F (200°C) and bake the bread for 30 minutes, until golden brown.
10. Remove the pan from the oven. Carefully firmly tap the pan onto a flat surface 2-3 times to release excess air from the bread. Remove the bread from the pan and cool completely on a wire rack.

Panna Cotta



By Edmund, Kye Been, and Sewen



Prep time 8 hours, including resting time



Preparation time 20-25 minutes



Serves 3–6

Summary

The Panna Cotta is a popular dessert, but with its popularity comes a variety of different recipes which are suggested online. Hence, we systematically studied the elasticity of Panna Cotta's made with different amounts of gelatin and milk-fat proportions. We find that using a medium amount of gelatin (1 tsp per 250ml of liquid) produces the best Panna Cotta. Using 1.5 tsp or more of gelatin makes for a product that is too firm, while using 0.5 tsp or less results in a Panna Cotta that can't hold its shape. We find that using dairy with a higher milk-fat percentage produces a richer end product as opposed to using dairy with a low milk-fat content like milk. We find that the combination that uses a medium amount of gelatin, and that uses heavy cream produces the best Panna Cotta.

Why this project

Inspired by the experiment on measuring the elasticity of the Vanilla Flan in the course, we wanted to explore another popular desert which has Italian roots, the Panna Cotta or “cooked cream” in Italian. A relatively simple desert, the Panna Cotta is generally made up of 4 components which

includes sugar, cream, vanilla, and gelatin. However, despite its simple components, it is difficult to perfect. A search online reveals a diversity of recipes which vary the amount of gelatin and types of dairy used — some incorporating milk or lower milk-fat creams while others do not — resulting in different textures. In addition, the Panna Cotta is a desert that does not require baking which makes it a very accessible yet delicious treat! As such, we are in search of the perfect Panna Cotta recipe to whip up in times of grave cravings and hope to share this wisdom with others.

Our experiment



Our preparations.

To find the ideal Panna Cotta ingredient combination, we experimented with two of the main components that make up the Panna Cotta. These

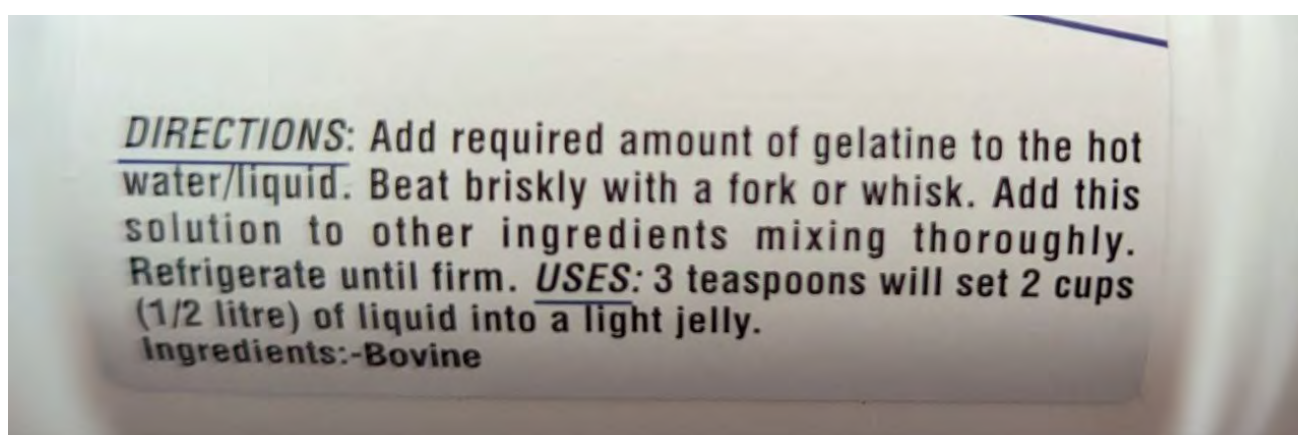
are the amount of gelatin and types of dairy used in the Panna Cotta. The sugar and vanilla extract amounts are kept constant.

We used a 3 by 3 matrix to reflect the different combinations of gelatin and creams used.

Cream	Cream (35% Fat)	Mix (50/50)	Light (Milk ~3%)
Gelatin			
Max (1 1/2 tsp)	Max/H	Max/X	Max/L
Med (1 tsp)	Med/H	Med/X	Med/L
Min (1/2 tsp)	Min/H	Min/X	Min/L

Table for the experiment matrix.

Procedure-wise, we began by measuring out the different amounts of gelatin and placing them in 3 different mixing bowls. In these bowls, cold water (1 tsp gelatin to 4 tsp water) was added and left for 10 minutes to allow the gelatin to bloom or re-hydrate. The maximum gelatin ratio was referenced from the recommended instructions that came with the gelatin power bottle.



Gelatine bottle instructions.

We used this ratio as a maximum reference point as it would be a given that

the company would want its customers to succeed in using their product. This ratio was 3 tsp gelatin to 500 ml of liquid. The calculations for a 250 ml portion of Panna Cotta mixture is shown below:

1. Max: 1000 ml to 6 tsp gelatin which is 250 ml to 1.5 tsp
2. Med: 1000 ml to 4 tsp gelatin which is 250 ml to 1 tsp
3. Min: 1000 ml to 2 tsp gelatin which is 250 ml to 0.5 tsp



Letting the Panna Cotta set in the fridge.

While the gelatin was blooming, we used an induction stove to heat a batch of cream (750 ml) to a simmer (80 degrees Celsius) along with the sugar and vanilla extract for 10 minutes to give time for the sugar to dissolve. Boiling the cream mixture was avoided as it is said to weaken the rigidity of the final product (gelatin begins to dissolve around 23 to 40 degrees

Celsius). Then, the heated mixture was poured into the 3 mixing bowls and stirred until the gelatin is fully dissolved. These mixtures were then poured into small containers, labeled, and left to set in the fridge overnight (gelatin sets at 15 degrees Celsius). This procedure was repeated 3 times for each cream type.

Our discovery



The elasticity measurements.

The following day, we measured the elasticity to find the ideal liquid-to-gelatin ratio that we can consistently rely on and how the different milk-fat variations impact the final texture of the pudding. We used a scale to measure the weight of a small object (a half-filled gelatin bottle) of 58 grams, which we used as a consistent weight for our experiment. We then

extract the Panna Cotta from the molds and measure its height before and after the weight is placed on it. We also observe the texture and consistency of each Panna Cotta variation through the highly nuanced taste test!

We find the following qualitative results:

Cream	Cream (35% Fat)	Mix (50/50)	Light (Milk ~3%)
Gelatin			
Max (1 1/2 tsp)	Max/H Very firm and thick consistency Which may be preferred for some	Max/X Relatively firm, similar to Max/H	Max/L Texture better - Lighter/silker because of the milk
Med (1 tsp)	Med/H Potentially the perfect combination Rich taste, while texture not too firm (maybe Max/L is a close second)	Med/X Could not hold shape Can't measure	Med/L Could not hold shape Can't measure
Min (1/2 tsp)	Min/H Can't hold shape but taste and texture actually pretty good if preference is soft texture. Like custard or the 'Lao Ban' soy pudding Similar to Med/X and Med/L	Min/X Liquid More viscous than Min/L	Min/L Liquid Sauce like consistency

The initial results.

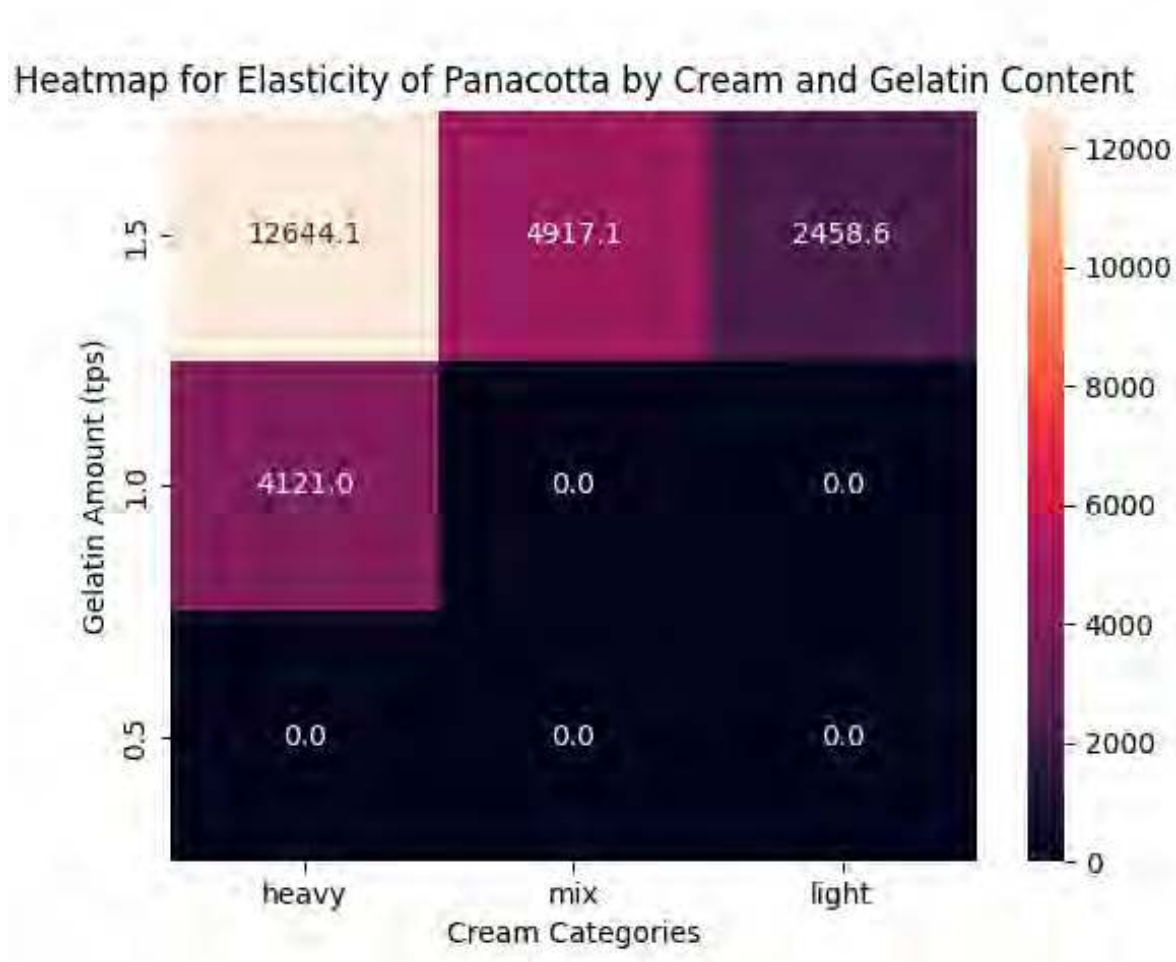
For the samples we can actual measure the elasticity, we quantify our findings of the Panna Cotta elasticity using the following equation:

$$E = \frac{(F \setminus A)}{(\Delta L \setminus L_0)}$$

Where

- F : the force exerted by the object we placed on the Panna Cotta bread calculated using its mass: $F = mg$.
- A : the surface area the force is in contact with.
- ΔL : the change in height of the Panna Cotta.
- L_0 : the initial height of the Panna Cotta.

We can visualize the results in the following heatmap:



The elasticity heatmap.

Why Gelatin Matters

The incorporation of gelatin gives the panna cotta a gel-like structure. At temperatures above 35°C, gelatin forms a solution with the panna cotta mixture, as the individual gelatin polymers are denatured and lose their regular triple helix structure. When the panna cotta mixture is cooled, gelatin polymers attempt to fold back into this regular arrangement, but this only happens partially. While different polymer strands are able to form bonds with each other, this new structure is irregular, and fluid molecules are trapped within it. As such, the milk/cream and water molecules are all contained within the gelatin, and this gives our panna cotta its custardy texture. The higher the gelatin content, the firmer the panna cotta as the polymer structure is denser due to more molecules being present. When the panna cotta is consumed, the gelatin is heated beyond 35°C and the gelatin polymer structure denatures once again, dissolving in our mouths for a rich, creamy experience.

The Role of Milk-Fat Content

On the other hand, the milk-fat content of the dairy used in making the Panna Cotta also has a role to play. For one, flavour molecules are carried in the fat molecules, and as such, the higher the fat content, the more flavour molecules can be found in the final product. Second, Fat affects not only the physical, oral thickness of the Panna Cotta, but its visual thickness as well. These factors play into the experience and appeal of the panna cotta, that in turn, can determine its desirability. Additionally, it reflects the role of the physicochemical properties of fat in our foods and the biological responses that feedback into our experiences on consumption. Lastly, fat has been found to play a role as a structural agent (at least in studies related to ice cream) which could partially explain why our Panna Cotta with a medium amount of gelatin and made only with heavy cream,

had a greater elasticity that our panna cotta made only with whole milk but the highest amount of gelatin.

Conclusion

In conclusion, gelatin is the biggest determining factor in the panna cotta holding it's shape while milk-fat plays a secondary role in affecting the sensory richness of the pudding.

As for the 'Perfect' panna cotta, we recommend either using only whole milk with maximum gelatin (1.5 tsp) or full cream with medium gelatin content (1 tsp). If you are looking for something more delicate, go with the whole milk option. If you want that creamy richness, go with the 35% milk-fat creams. Since cost can be a factor, the milk recipe is more affordable so we will include that recipe below.

But before we get to the recipe of perfection, we would like to leave some food for thought, for future Panna Cotta aficionados. How might varying the amount of sugar affect the structure of the Panna Cotta? Alternatively, how might the use of Agar-Agar (a hydrocolloid extracted from red seaweeds) result in a different kind of Panna Cotta?

Ingredients

☀ 4 tbsp of sugar ☀ 2 tsp of vanilla extract ☀ 250 ml of milk ☀ 1.5 tsp of gelatin ☀ 6 tsp of water

Equipment

☀ Induction Cooker ☀ Small pot

Directions

1. Measure 1.5 tsp of gelatin and put into a mixing bowl.

2. Add 6 tsp of water into the gelatin and let it bloom for 10 minutes.
3. Use an induction stove to heat up the milk (250 ml) to a simmer (80 degrees celsius) along with the sugar (4 tbsp) and vanilla extract (2 tsp) for 10 mins, to give time for the sugar to dissolve. Do not let the mixture boil.
4. Pour the heated mixture into the mixing bowl and stir until all the gelatin is fully dissolved.
5. Pour the mixture into small containers, and leave to set in the fridge overnight (gelatin sets at 15 degrees celsius).



The finished Panna Cotta!

Cream Puffs



By Charis, Kristen, and Yi Qian



Preparation time (1 batch) 30 minutes



Serves 12 cream puffs

Summary

In this experiment, we explored the scientific basis behind cream puffs. Specifically, we investigated whether the protein content in the flour used affects the quality of cream puffs made, in terms of structure and mouth-feel. This will determine whether it is warranted to use bread flour instead of other types of flour that home bakers typically use, such as plain flour. Thus, in our experimental method, we varied the protein content of each batch of cream puffs by varying the type of flour used (cake flour, plain flour, and bread flour). Overall, we found that cream puffs made with bread flour (i.e., the flour with the highest protein content) resulted in the highest weight loss, highest rise, and highest cavity percentage in the cream puffs. In other words, using bread flour to make cream puffs is warranted as it produces the lightest, airiest, and creamiest cream puffs.

Background

Choux pastry, or *pâte à choux*, is a dough used in many pastries, including cream puffs. Choux pastries are irregular cabbage-like balls with a hollow center and are characterized by a crispy outer shell with a light, airy interior. Choux pastry is interesting because, unlike other pastry doughs, it

is cooked twice: once to prepare the paste itself and once again to create hollow puffs.

Why this project

We were looking at a cream puff recipe in the cookbook *baker & cook* by Dean Brettschneider and found that the recipe called for bread flour. We were curious as to whether there will be a significant difference in the quality of the cream puffs to warrant using bread flour as opposed to other types of flour. This is because most home bakers might only use or have plain flour.

Our research question came to be: *Does the protein content in the flour affects the quality of cream puffs made, in terms of structure and mouthfeel?* Our hypothesis is: The higher the flour's protein content, the higher the cream puff will rise, the lighter the cream puff will be, and the better the mouthfeel will be.

Methodology

We baked three batches of cream puffs, varying the protein content of each batch by varying the type of flour used. Namely, we used cake flour, plain flour, and bread flour. Different flour brands will have slightly different protein contents, but for the brands we used, the protein contents were 7.8, 10, and 13 percent for the cake, plain, and bread flours, respectively. To prepare the cream puffs, we followed the recipe below.

Ingredients

For the choux pastry (Makes 3-4 cream puffs)

☼ 25mL water ☼ 25mL milk ☼ 22.5g salted butter ☼ 34g flour (cake, plain, and bread flour) ☼ 50g eggs (about 1 small egg)

For the filling (Makes enough for 12 cream puffs)

✻ 240mL milk ✻ ¼ tsp salt ✻ 60g sugar ✻ 1 vanilla bean or 1 tbsp vanilla bean paste ✻ 2 egg yolks ✻ 25g cornstarch ✻ 180g whipping cream

Equipment

✻ Small saucepan ✻ Piping bags ✻ Large round piping tip (optional)
✻ Oven

Directions

For the choux pastry:

1. Preheat the oven to 200°C. Line a sheet tray with parchment paper. Using a circle cookie cutter, draw four circles spaced apart on the parchment paper. Flip the parchment paper over and set the sheet tray aside.
2. In a small pot over medium heat, heat the water, milk, and butter until boiling.
3. Once boiling, remove from the heat and add the flour (starting first with cake flour), stirring well with a wooden spoon to combine. The texture of the dough will resemble mashed potatoes.
4. Return the pot to heat and cook the dough for 4 minutes, stirring constantly. A layer of film will develop on the bottom and sides of the pot as the dough is cooking.
5. Allow the dough to cool for about 5 minutes. Meanwhile, beat the egg in a separate bowl.
6. When the dough is warm but not hot to the touch, add the egg in small additions, whisking after each addition. Add enough egg such that the

dough holds soft peaks. Then, transfer the dough to a piping bag fitted with a large round tip.

7. Pipe the dough onto the sheet tray, using the circles drawn earlier as a size guide.
8. Bake for 15 minutes, ensuring never to open the oven door before the 15-minute mark.
9. Remove from the oven and place on a wire rack to cool. Weigh and measure the height of each cream puff. Cut the cream puffs in half vertically and measure the size of the cavity.
10. Set the cream puffs aside and repeat with the remaining two types of flour.

For the filling:

1. In a small saucepan, heat the milk, salt, sugar, and vanilla over low heat.
2. In a separate bowl, whisk the egg yolks with cornstarch.
3. When the milk just begins to steam, turn off the heat. Add $\frac{1}{4}$ cup of the milk mixture to the egg yolks, whisking immediately.
4. Combine the egg mixture with the milk mixture in the saucepan.
5. Return the saucepan to low heat, whisking constantly. Cook the mixture until it is thick enough to coat the back of the wooden spoon.
6. Transfer the custard to a container. Cover the top with clingfilm, ensuring that the clingfilm touches the surface of the custard. Allow to cool to room temperature then transfer to the refrigerator to cool completely.

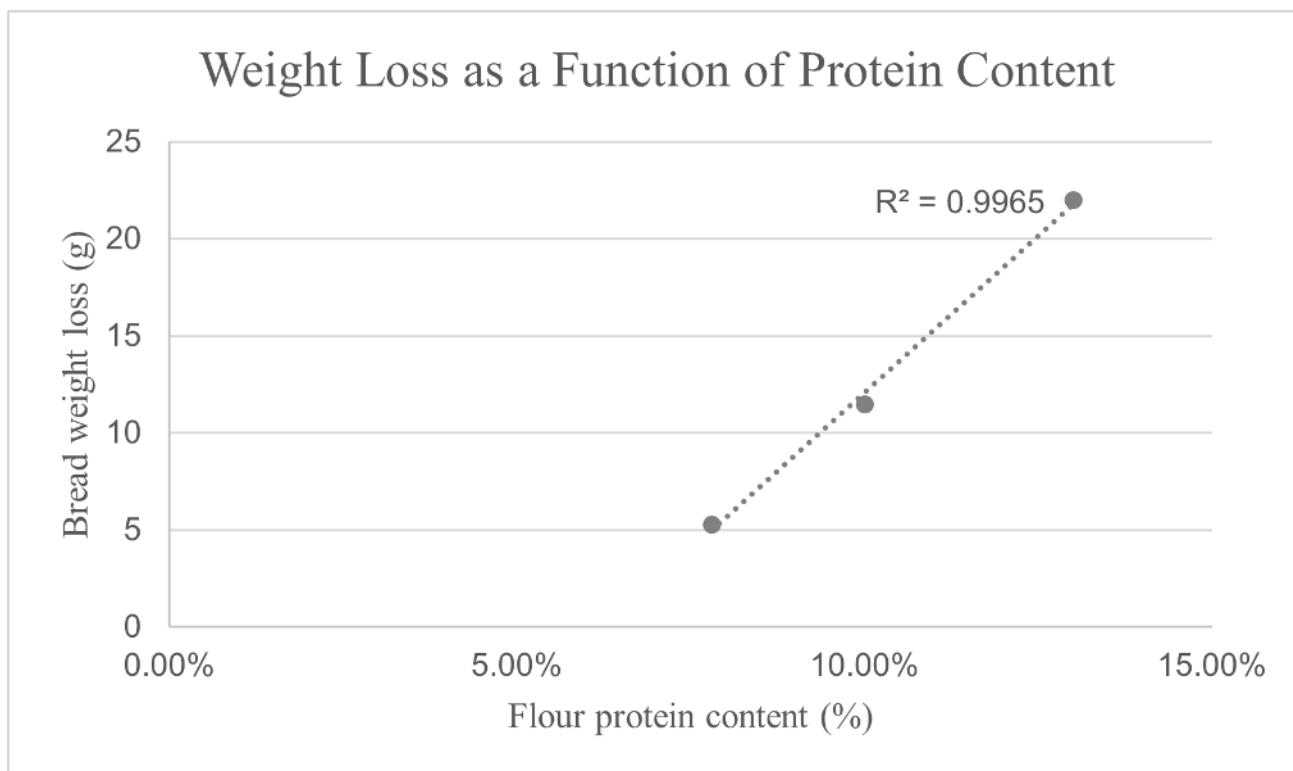
7. When ready to fill the choux pastry, whip the whipped cream to soft peaks in a large bowl. Add the cooled custard to the whipped cream and whip to combine.
8. Transfer to a piping bag fitted with a small round tip. To fill the pastries, make a small incision in the bottom of the pastries and fit the piping tip into the incision. Fill and enjoy your cream puffs!



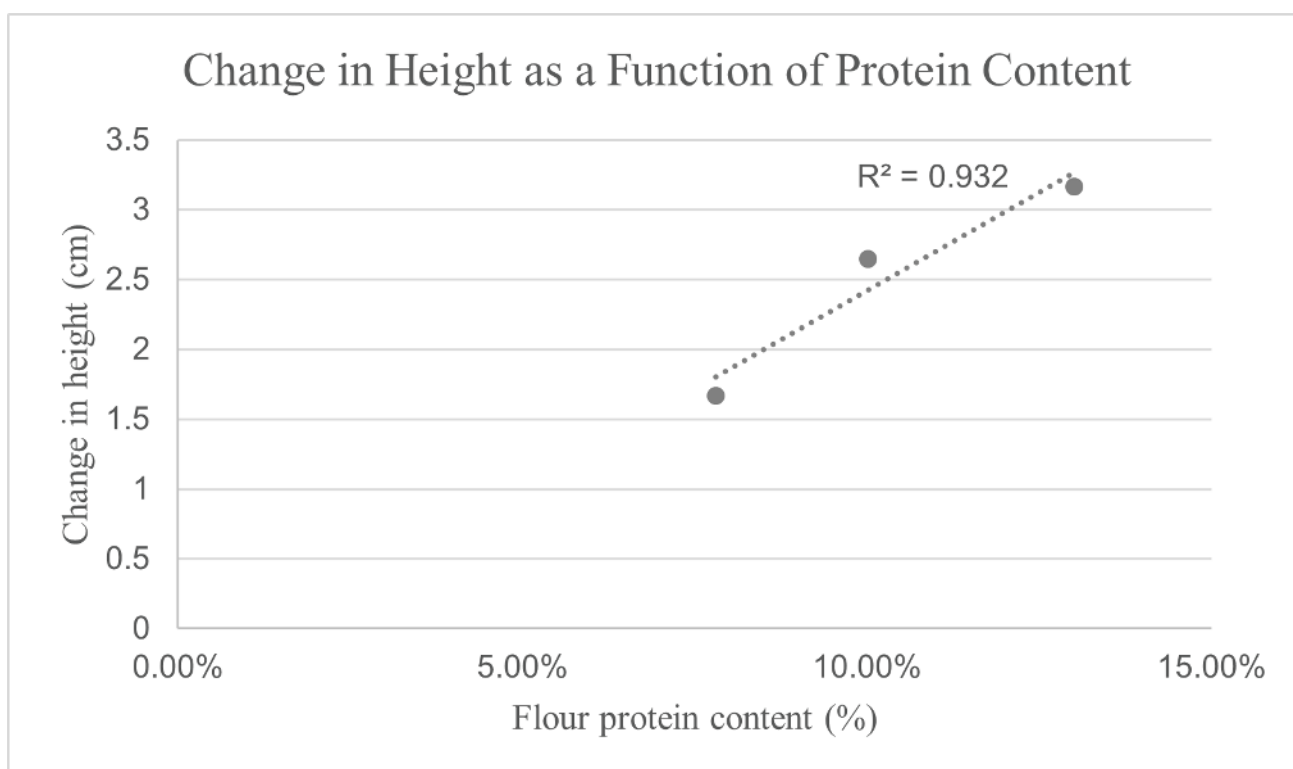
Cream puffs with matcha and custard filling

Our Discovery

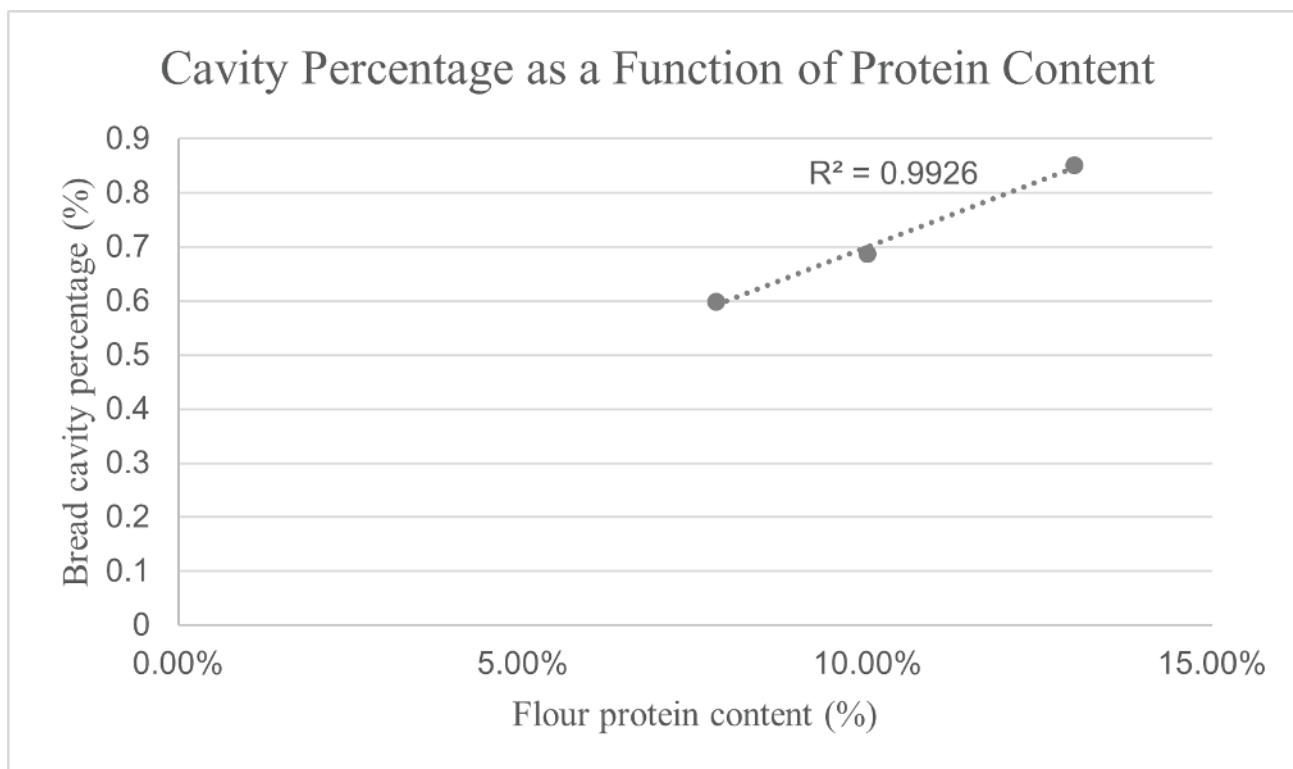
In our experiments, we found the following trends: (1) bread weight loss (i.e., the bread's airiness and lightness) **linearly increases** with the flour's protein content, (2) change in height (i.e., the bread's ability to rise) **linearly increases** with the flour's protein content, and (3) the bread's cavity percentage **linearly increases** with the flour's protein content. These are shown in the graphs below:



Bread Weight Loss as a Function of Protein Content



Change in Height as a Function of Protein Content



Cavity Percentage as a Function of Protein Content



Cavity of the cream puff made with cake flour (bottom) and plain flour (top)



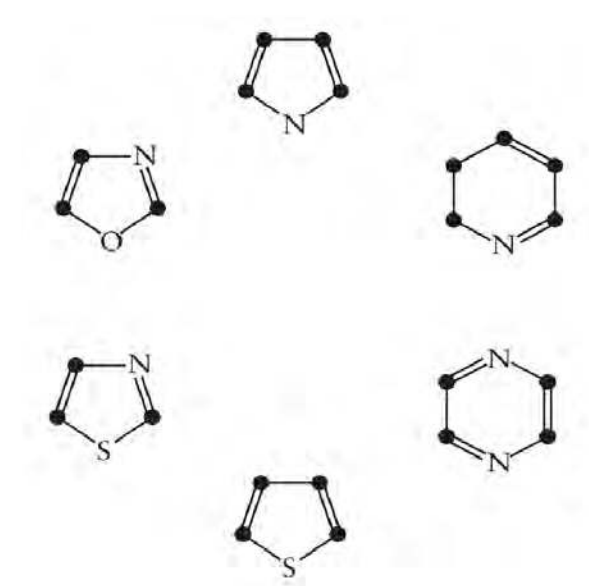
Cavity of cream puff made with bread flour

The Science Behind Our Results

Role of the control ingredients in baking the cream puffs

Eggs play three roles in baking the cream puffs: (1) they act as leavening agents where the fat in the yolks gives baked goods a delicate, fluffy feel, (2) they also serve as an emulsifier to give the cream puffs a uniform and smooth texture, and lastly, and (3) they act as an extra source of proteins, where the cream puff's structure is strengthened by the proteins.

The **fats** in the butter and the milk play a role as shortening agents. Fats and oils “shorten” a dough or weaken its structure, making the final product more tender or flaky. This role is apparent, especially in puff pastry, where layers of solid fat divide thin layers of dough from one another into distinct layers of pastry. However, it must be noted that the right amount of fat should be used, or the fats will interfere with gluten production and the cream puffs will collapse.

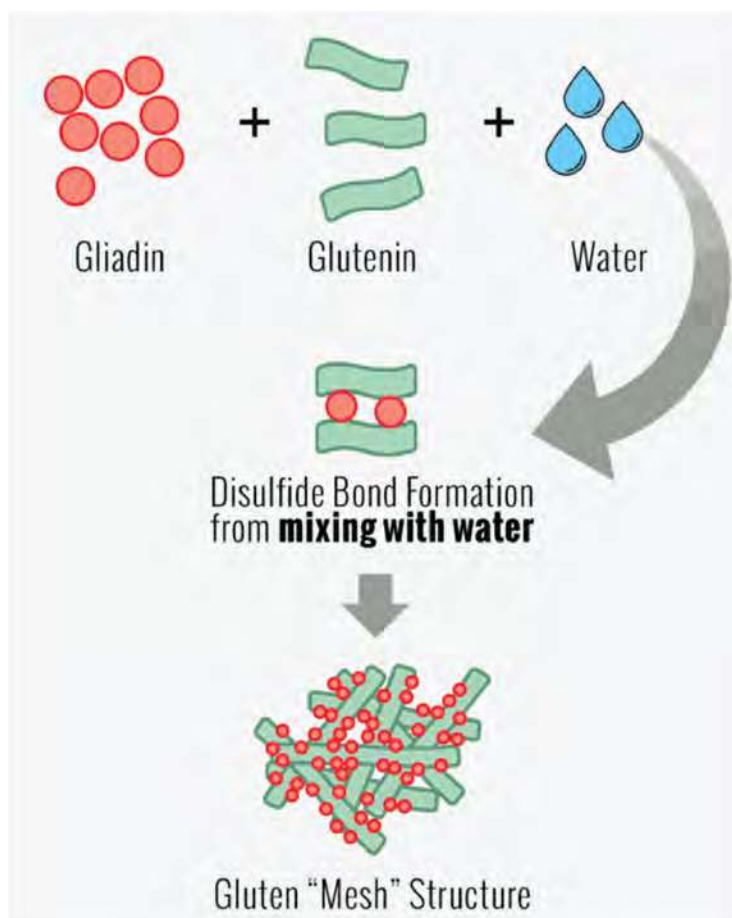


Representative molecules produced by the Maillard reactions between carbohydrates and amino acids
(McGee, 2004)

Heat contributes to the **Maillard reaction**. The Maillard reaction is responsible for the cream puff's brown crust. The reaction between an amino acid and a carbohydrate molecule (either free or linked to starch, glucose, and fructose) is more reactive than table sugar, which initiates the series. An unstable intermediate structure is then created, which goes through additional modifications resulting in hundreds of distinct by-products. The unique central rings of pyrroles, pyridines, pyrazines, thiophenes, thiazoles, and oxazoles are partly produced by nitrogen and sulfur atoms from amino acids. It is possible to add additional structures to the carbon atoms in each ring.

In simpler words, a chemical interaction between amino acids (proteins) and carbohydrates (sugars) causes our crust to brown when it is heated to a minimum of 280°F. This indicates that the starches in our food are becoming simple sugars and caramelizing. Additionally, the gluten matrix's

proteins are broken down into simpler amino acids, which recombine with other molecules in our dough.

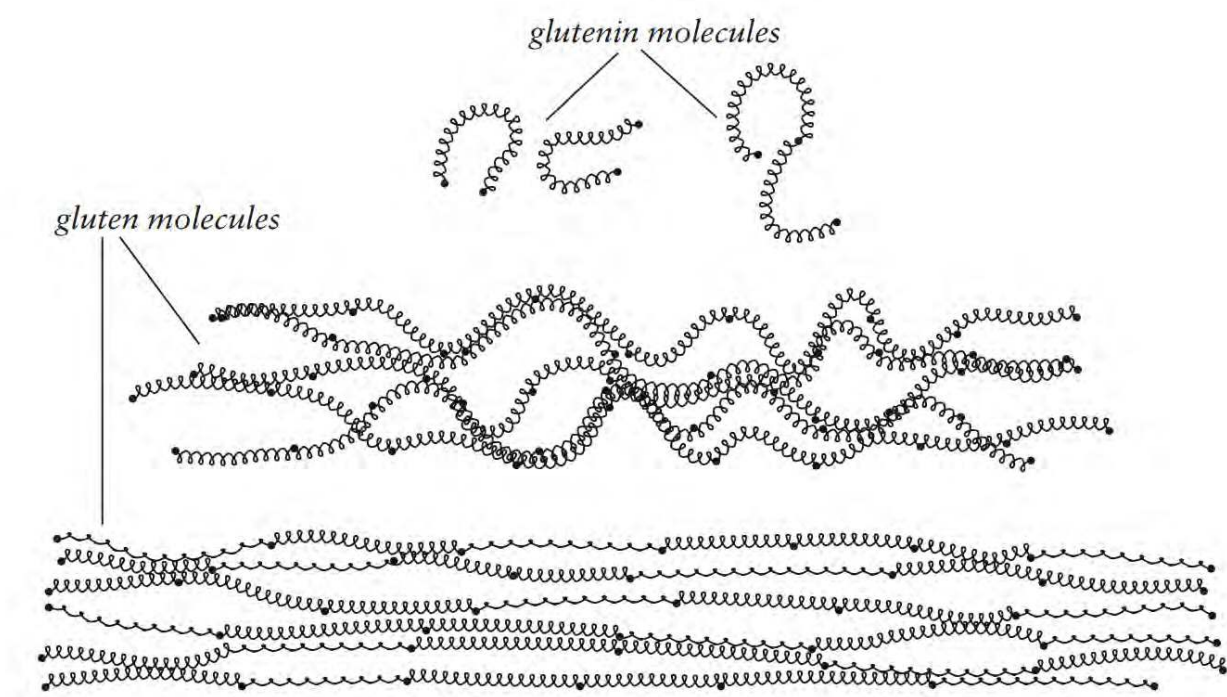


Process of gluten formation (Gavin, 2022)

Overall role of flour in baking cream puffs

Higher protein content in the flour makes for **stronger gluten**. Gluten is a complicated mixture of certain wheat proteins that do not dissolve in water but do associate with water molecules and with one another. The proteins are immobile and inactive when they are dry. However, they have the ability to alter their shape, move in relation to one another, and create and break bonds when exposed to water. The gliadins and glutenins, which make up the majority of the gluten proteins, have a length of roughly 1,000 amino acids. The glutenin proteins and the gliadin chains form a compact mass once the gliadin chains fold over one another. The

glutenins, however, interact with one another in a variety of ways to create a vast, closely-knit network. Sulfur-containing amino acids can create potent sulfur-sulfur interactions with the same amino acids at the ends of other glutenin chains at either end of the glutenin chain. They need the presence of oxidizing agents for this to happen, such as airborne oxygen, specific compounds made by yeasts, or “dough improvers” that are added by the flour producer or baker.



Gluten and glutenin molecules (McGee, 2004)

The majority of the long, coiled middle portion of the glutenin molecule is made up of amino acids that associate with other amino acids to generate weaker, transitory connections (hydrogen and hydrophobic bonds). In order to create super-chains of a few hundred glutenins, glutenin chains connect up end to end. Coiled stretches along these super chain lengths easily build numerous temporary linkages with corresponding stretches along nearby gluten proteins. As a result, a large, linked network of coiled proteins is created.

Due to their compact nature, the gliadin proteins act as “ball bearings” for the glutenins, allowing some of them to slide past one another without connecting, which is what causes gluten flexibility. The linked gluten proteins’ coiled and kinked structures give rise to elastic and gel-like properties of raw dough.

Role of the flour in the cream puff’s structure and mouthfeel

Higher protein content in the flour gives the cream puffs a **higher rise**. Here, gluten forms a stretchy framework that traps the carbon dioxide inside the dough, making the dough rise more vertically as opposed to spreading out sideways. On the other hand, low-protein wheat produces weaker gluten that is less able to hold carbon dioxide, resulting in bread that is more cake-like and flatter. This type of flour works well for light sponges, biscuits, cookies, scones, soda breads, and other chemically leavened baked goods. However, it does not work so well with cream puffs.

Higher protein content in the flour makes the cream puffs **lighter**. In the oven, the bread releases moisture as it is baking. Because water is a component of the bread dough, whether it may be in the milk, butter, or just water added, the water evaporates as it is baking and the weight of the bread gets reduced. Dough made with high-protein flour promotes more water loss as more water evaporates when the bread rises more. Thus, this leads to more weight loss in the bread.

Finally, **higher protein content** in the flour makes for **hollower cavities** in the cream puff. This is also due to the higher rise, where the inner network or layers of the dough gets stretched upwards. We are able to see this in figures 4 and 5, where (a) the cream puff made with cake flour still has layers that are not well separated in its cavity, (b) the cream puff made with plain flour has a slightly improved cavity, and (c) the cream puff made with the bread flour has the most developed and hollow cavity,

perfect for piping the cream in.

Limitations and Improvements for Future Research

To vary the protein content, we used different types of flour. We recognize that in using different brands and types of flour, there might be other factors that are playing a role in varying the quality of the cream puffs. As a potential improvement for future research, we recommend using one type of flour (cake flour or plain flour) and adding varying amounts of gluten powder (also known as vital wheat gluten) to simulate different protein contents in the flour.



From left to right: Creampuffs made with bread flour, plain flour, and cake flour

Conclusion

Overall, through the experiments we carried out, we found that cream puffs made with **bread flour** (i.e. the flour with the highest protein content) resulted in the **highest weight loss, highest rise, and highest cavity percentage** in the cream puffs. In other words, the use of bread flour in making cream puffs is warranted as it produces the lightest, airiest, and creamiest cream puffs. These were followed by the cream puffs made with plain flour, and finally, the cream puffs made with cake flour. As an area for future research, it might be worth investigating how varying the amounts of other ingredients in the cream puffs (such as the eggs, milk, or butter) can also affect the overall quality of the puffs.



Cream puffs with matcha and custard filling

Matcha Basque Cheesecake



By Yan, Carine, Amanda



Preparation time 5-10 minutes



Baking time 25 minutes



Serves 1

Summary

Loved by many, matcha basque cheesecake is a comfort dessert that couples the aroma of matcha with the thick consistency of a creamy filling. This dish follows an easy recipe, needs few ingredients, and only an air-fryer, making it suitable for college students to make in their dorms. However, there are many ways a cheesecake can go wrong. It could be too dense, dry, soft, liquidy, or acidic when the wrong proportion of ingredients is added. We investigated the effects of adding different levels of each main ingredient in our recipe, thereby varying the concentration of saturated fat, unsaturated fat, and carbohydrates in the cheesecake batter. Our findings showed adding more heavy cream and sugar together gives a less acidic and fine-textured cheesecake, suggesting that you can reproduce the recipe with specific ingredients according to their properties and what you prefer regarding texture and taste.

Why this project

We were initially inspired by Keong Saik Bakery, which is known for its matcha basque cheesecake that is creamy and smooth in texture, and rich

in flavor. We decided to work on this in our project as we wanted to recreate an easy and beginner-friendly dessert we could easily make in our dorms. We were also curious about what made a good cheesecake, given that it can be quite subjective. Therefore, we wanted to experiment with adding different proportions of ingredients to the existing recipe to understand how each ingredient can affect the taste or texture of the cheesecake. We can then learn which ingredient in the recipe to alter according to our preferences.

In addition, we chose a recipe for baking mini-sized cheesecake for a couple of reasons. One, it is convenient because it is a perfect bite-sized snack that is easier to eat on the go — good for busy college students. This saves the hassle of cutting and storing a whole large cheesecake, especially since most college students lack the storage space or the proper cutting materials for this. Two, the smaller size means multiple cheesecakes can fit inside a typical air fryer at any one time, creating more cheesecakes for people to enjoy. Additionally, the small size requires less baking time to bake the batter all the way through, significantly cutting down the time needed to implement this recipe. Lastly, considering how dense cheesecakes can be, the smaller size makes them less greasy and more delectable!

Our experiment

Our ingredients were obtained to best replicate the ingredients used by the control recipe. We got Philadelphia cream cheese, 35% fat whipping cream, fine sugar, eggs, matcha powder, and all-purpose flour. We made four different batters, one following the control recipe and three others with two times the stated amount of sugar, heavy cream, and cream cheese, respectively. Using aluminum foil tarts, we baked the four different cheesecake batters simultaneously in the air fryer at 200°C for 25 minutes, which is almost half the time it takes to bake in the oven. As

the air fryer cooks faster than a regular convection oven due to the super-convection and smaller cooking space, we figured we would need much less cooking time. We assessed our results on the final appearance and taste based on their fluffiness (aerated nature) and the condition of the burnt top.

Our discovery



Our control and experimental cheesecakes

Control recipe macronutrients breakdown:

Ingredients	Quantity (g)	Carb(g)	Protein(g)	Fats(g)	Water(g)
All-purpose flour	0.975	0.731	0.244	0.00936	0
Egg	7.27	0	0.909	1.06	5.60
Granulated Sugar	12.5	12.5	0	0	0
Whipping Cream	7.5	0.233	0.15	2.63	5.03
Cream Cheese	56.3	0	4.0	20.1	0
Total Weight (g)	84.5	13.5	5.30	23.8	10.63
Percentage Weight(%)	-	15.9%	6%	28.1%	12.5%

The control looked the smoothest considering both its burn and inner texture. The acidity from the cream cheese was present but not too overpowering, and the matcha flavor came out the most prominently compared to the batters with changed variables. The sugar serves to create tenderness and moisture and balances out the acidity of the cream cheese, the

heavy cream serves to add fluffiness to the cheesecake, the cream cheese creates the creamy texture of the cheesecake, and the flour serves as a starch that prevents the cake from cracking and the egg proteins from over-coagulating.

2X Sugar Recipe Macronutrients Breakdown:

Ingredients	Quantity (g)	Carb(g)	Protein(g)	Fats(g)	Water(g)
All-purpose flour	0.975	0.731	0.244	0.00936	0
Egg	7.27	0	0.909	1.06	5.60
Granulated Sugar	25	25	0	0	0
Whipping Cream	7.5	0.233	0.15	2.63	5.03
Cream Cheese	56.3	0	4.0	20.1	0
Total Weight (g)	97	26	5.30	23.8	10.63
Percentage Weight(%)	-	26.8%	5.46%	24.5%	11%

For the cheesecake with 2x sugar, the burn at its top tasted slightly more caramelized due to the sugar and had the smoothest interior due to the increased viscosity from the extra sugar. As sugar interferes with protein coagulation, it raises the temperature for bonds between the flour and eggs to form, creating softness and tenderness within the batter (although too much sugar could break the structure and make the cake too crumbly!). Sugar also creates more viscosity which bulks the components of the batter together and tends to hold onto water molecules (hygroscopic) and keep the batter moist. These factors may explain the smoothness and moistness of this cheesecake. In addition, this cheesecake tasted the least acidic due to the sugar overpowering and balancing the acidity of the cream cheese.

2X Whipping Cream Recipe Macronutrients Breakdown:

Ingredients	Quantity (g)	Carb(g)	Protein(g)	Fats(g)	Water(g)
All-purpose flour	0.975	0.731	0.244	0.00936	0
Egg	7.27	0	0.909	1.06	5.60
Granulated Sugar	12.5	12.5	0	0	0
Whipping Cream	15	0.466	0.3	5.26	10.1
Cream Cheese	56.3	0	4.0	20.1	0
Total Weight (g)	92	13.5	5.45	26.4	15.7
Percentage Weight(%)	-	14.7%	5.92%	28.7%	17%

For the cheesecake with 2x whipping cream, the batter was a lot more liquid throughout the baking process due to the extra liquid from the heavy cream. The result tasted the most fluffy, or aerated, due to whipping cream allowing more gas bubbles to enter the batter during the whipping process. The burnt top also appeared the most even and most brown. The whipping cream we used contains 35% fat and is a liquid. We calculated the percentage weights of the ingredients and found that this recipe has a higher fat and water content than the control recipe. Our results suggest that adding more whipping cream to the batter adds tenderness and flavor from the fat and moisture from the water content, softening the texture of the cream cheese. When folding the whipping cream as we were mixing the cake batter, some of the fat globules that formed the semisolids were damaged in the process. We found that adding two times more whipping cream resulted in a fluffier, more aerated, and well-mixed texture and a more evenly visually appealing burnt top compared to the other cheesecakes.

X2 Cream Cheese Recipe Macronutrients Breakdown:

Ingredients	Quantity (g)	Carb(g)	Protein(g)	Fats(g)	Water(g)
All-purpose flour	0.975	0.731	0.244	0.00936	0
Egg	7.27	0	0.909	1.06	5.60
Granulated Sugar	12.5	12.5	0	0	0
Whipping Cream	7.5	0.233	0.15	2.63	5.03
Cream Cheese	112.6	0	8.0	40.2	0
Total Weight (g)	140.8	13.5	9.30	43.9	10.63
Percentage Weight(%)	-	9.59%	6.61%	31.2%	7.55%

For the cheesecake with 2x cream cheese, the batter was very thick and could not be adequately mixed. Lumps of cream cheese remained even after stirring for a long time. After baking, the surface of the cheesecake was the most uneven; some parts were very burnt, while others remained green. The cake was very heavy, and the texture was very dense. It was not smooth and tasted very heavy. Lastly, it was also the most acidic due to the acidity of the cream cheese overpowering all other flavors from the sugar, matcha, and other ingredients.

The mixture is dense and heavy due to the increased percentage of fats and protein caused by the addition of more cream cheese. As shown in our calculations, compared to the control recipe, the recipe for 2x cream cheese contains a noticeably higher percentage of fats and protein. Furthermore, due to the lower percentage of water, the increased amount of protein in the batter (as shown by the higher percentage of protein) coagulates with the fat globules, making the texture thicker and denser. Because the mixture is denser, it is unevenly exposed to heat in the air fryer; therefore, the browning area on the cheesecake is uneven. As for acidity, it comes from the acid used in the cheese fermentation process for cream cheese. In the control recipe, with a moderate amount of cream cheese, the tartness

of the cheese itself is balanced out by the sugar, matcha, and other taste molecules in the mixture. However, 2x cream cheese has significantly more cream cheese than any other ingredient; therefore, the sour acidic taste becomes overpowering. Furthermore, since the batter mixture for 2x cream cheese was the thickest and not easy to mix, the acidity could also result from the cream cheese clumps that could not be mixed in.

Limitations: Although we controlled variables and ensure that only one component of the cheesecake recipe was being tested at any one time, there are some limitations to our study. For example, in the recipe we used, the ingredient portions are divided into eight. While we used measuring spoons, there is still a human error in these measurements. Moreover, though we controlled for temperature variations by baking all our cheesecakes at the same time, there is a possibility that the airflow and heat distribution in the air fryer is uneven.

Conclusion and lingering questions

In conclusion, we found that we could make a good matcha basque burnt cheesecake if we change the number of ingredients one adds to a baking recipe according to our preferences and understand what each ingredient does. The best cheesecake recipe remains subjective as some found the recipe with more sugar to be tastier while some preferred the one with more whipping cream. We discovered that changing the proportion of fat, protein, and liquid by adding more whipping cream and sugar could significantly change the texture and taste of the cheesecake. We decided that the best combination for a burnt matcha basque cheesecake would be to add 1.5 times more sugar and 1.5 times more whipping cream to the original recipe to perfectly neutralize the acidic taste of the cream cheese, give a smoother, fluffier texture, and produce a more evenly burnt top. In the future, we could also explore the effect of increasing or decreasing one

component of the recipe, or how using different types of flour affects the texture.



Mini matcha basque cheesecake!

Ingredients

✧ ¼ cup of Philadelphia cream cheese ✧ 1.5 tbsp of white sugar (compared to 1 tbsp in the original recipe) ✧ 0.75 tbsp of heavy whipping cream (compared to 0.5 tbsp in the original recipe) ✧ 7.5 mL egg ✧ Approx ⅓ tsp all-purpose flour ✧ Pinch of salt ✧ 1 tsp of matcha powder

Equipment

✧ Measuring Spoons and Bowl ✧ Aluminum Foil Cupcake Baking Cup ✧ Air Fryer

Directions

1. Preheat the oven to 400°F (200°C).
2. Mix together cream cheese, egg, and heavy cream in a mixing bowl.
3. Mix together the dry ingredients: sugar, flour, matcha, and a pinch of salt.
4. Mix until well blended and creamy.
5. Fill in the aluminum baking cup with the batter evenly, and bang the cup lightly on the table to remove excess air bubbles.
6. Place in the air fryer for 25 minutes. The cheesecake is ready when the top is brown.
7. Take the cheesecake out to cool for a few minutes, and enjoy!

Mochi Donuts



By Dabeer, Gunnar, and Rachel



Prep time 15 minutes



Cook time 15 minutes



Serves 10–12

Summary

The quality of a mochi donut is determined by its chewiness, which can be modified by adjusting the proportion of sticky rice flour in the recipe. This is because glutinous rice flour mainly consists of amylopectin — a branched starch molecule that forms crosslinks with other molecules after being heated, causing mochi donuts to be elastic after being fried. Hence, by modifying an existing mochi donut recipe, our group seeks to discover an ideal proportion of rice flour to create the best mochi donuts. We also seek to provide a range of sticky rice flour proportions for people to adjust according to their texture preference while ensuring that the baking process is smooth (the dough can be rolled and shaped). The ideal chewiness is determined by measuring the elasticity of the donut and the preferences of 13 taste-testers

Why this project

In the baking workshop, we learned how gluten formed by proteins in bread flour gives it a chewy texture. This piqued our interest in mochi donuts — even though it is gluten-free, it has a chewy texture. Through

this experiment, we hope to discover how elasticity in starch-based foods is created and how much the proportion of starch affects the process of creating and the textures of these foods!

Our experiment

Using the recipe cited below, we prepared three different groups of donuts: a recipe-following control group that used 225g of rice flour and experimental groups with 250g and 275g rice flour. We standardized mixing times at 4 minutes, frying time at 5 minutes, and cooling times at 2 minutes, used a dough hook at low speed for all mixing, and cut the donuts into 25g, 1cm thick disks. After cooling, we used force-displacement measurements with 400, 600, and 800g weights to determine the elastic modulus of the cooked mochi donuts. We did this by measuring a bisected donut and using a scale, ruler, and fixed weights to measure the difference in height. Elastic modulus was calculated using Young's formula for donuts in each group.

$$\Upsilon = \frac{F/A}{\Delta L/L} = \frac{\text{Stress}}{\text{Strain}}$$

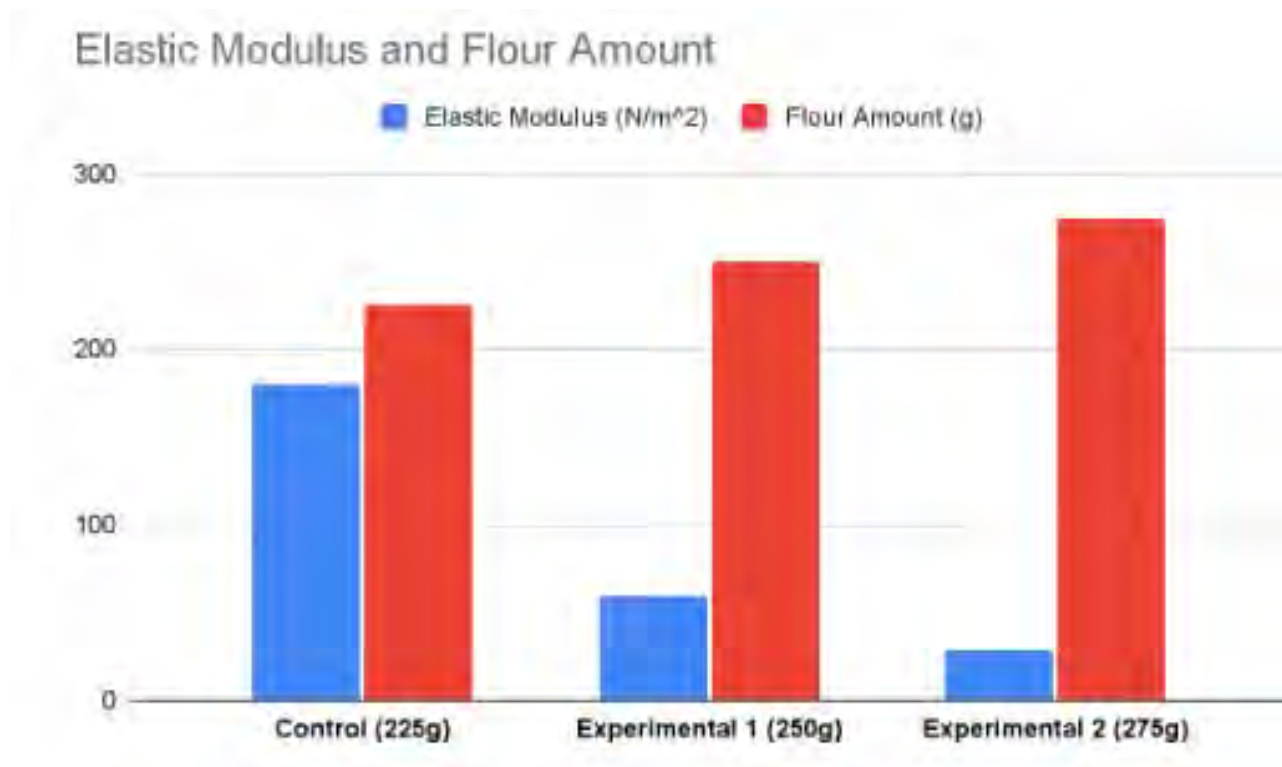
The measurements were then averaged and compared. In addition to this, we had a sample size of 13 compare the three donuts for mouthfeel, answering questions on chewiness, texture, and overall desirability.

Our discovery

Using our two different methods, we found that the flour proportion in the dough is inversely proportional to the donuts' elasticity and "softness".

By increasing the flour in the dough by 11% from the control, we found that the elasticity significantly decreased from 180 N/m² to 60 N/m². Done again, increasing the flour proportion by an additional 10% (25g) yielded a

28.5 N/m² elastic modulus. This translates to an increasingly brittle structure in the donuts, starting with quite chewy and turning into a harder and consequently less chewy mass. The exponential decline in elasticity might be explained by the unique bonding structure and patterns of amylose and amylocine, of which little public research is available. This may look like a mixing method that stretches and entangles these starches or perhaps the inhibiting behavior of excess reagent rice flour during the cooking phase, which prohibits crosslinks from being formed. This second reason is what we suspect caused the sharp decline in elasticity.



By subjective standards, people found a similar liking to the control (225g) and experimental 1 (250g) batches of donuts. The control was mentioned to be softer than experimental 1 (250g), and for those that could tell the difference between the two, they preferred experimental 1 (250g). The experimental 2 (275g) donuts were notably harder and more brittle than the first two batches. Because of this, we recommend changing the recipe

to 250g of flour dough while acknowledging it is essentially a matter of taste.

Summary and lingering questions

To conclude, we found that using more sticky rice flour results in a harder dough and decreases the elasticity of the mochi donut. By trying 225g, 250g, and 275g, we yielded, through testing, that people mostly preferred donuts with 250g of rice flour, having the ideal mouthfeel and chewiness.

For future prospects, we aim to explore mochi donuts with different types of rice flour with different protein content. This will also have an effect on the elasticity and mouthfeel, and then we aim to compare the best elasticity and mouthfeel of each flour we experiment with our donut with.

Ingredients

Starter dough

✧ 30g sticky rice flour ✧ 31g of whole milk

Main dough

✧ 250g of sticky rice flour (225g if you prefer your donuts soft, 275g if you prefer it more chewy) ✧ 120g of whole milk ✧ 35g of unsalted butter, melted ✧ 50g of granulated sugar ✧ 1 egg ✧ 1 tsp of baking powder

Coating

✧ 5g of Sugar ✧ 1 tsp Cinnamon Powder

Equipments

✧ Microwave ✧ Microwave-proof bowls ✧ frying pot ✧ Stand mixer

bowl with dough hook (optional) ✨ Cooling rack ✨ Dough cutter

Directions:

1. To make the starter dough, mix the sticky rice flour and whole milk. Microwave the mixture for 30 seconds. It should look opaque and bouncy. Set it to cool.
2. Meanwhile, for the main dough, mix all the ingredients together until combined. You may use your hands or a stand-mixer bowl with a dough hook. Add in your starter dough into the main dough, and continue mixing until a dough mass has formed and the starter dough has completely blended into the mixture. You should be able to touch the dough without it sticking to your finger.
3. Scrape the dough onto a surface that is dusted with glutinous rice flour. Sprinkle some flour on the dough and your hands, for an easier shaping process.
4. Roll the dough to about 1.3cm thick. Use a cutter to cut out as many donuts as you can
5. Add enough oil to a frying pot (depending on the pot size, the amount of oil should be enough to fully submerge the donuts), and set it on medium-high to bring the oil to 165°C (if you do not have a thermometer, place a wooden chopstick in the oil, if small bubbles form around it quickly, the oil is ready).
6. Without crowding the pot, carefully place a few donuts into the oil. Fry for 2 minutes on each side (total 4 minutes/ until golden brown). Initially, it will sink to the bottom, but it will soon puff up, turn golden brown and float to the surface.

7. Transfer to a cooling rack lined with paper towels to drain out the oil.
8. Meanwhile, mix the sugar and cinnamon powder on a plate. Once cooled, coat the donuts with the sugar-cinnamon mixture and enjoy! (You can also enjoy these donuts with any other toppings e.g. icing!)



The Mochi Donuts: The Easiest treat you can make!

Apple Crumble



By Raphael, Aaron, Silky, and Zirdi



Prep time 30 minutes



Baking time 30 minutes



Serves 12

Summary

Apple crumble is a simple-sounding dessert with two main components: the apple and the crumble. That being said, a lot can go wrong in the making of this dessert. For this final project, our group tackled two main problems within the main components of an apple crumble. Namely,

1. *In the crumble: How do we determine the types of ingredients (sugar, flour) that optimize the crispiness or crunchiness of the crumble?* We kept the proportion of ingredients constant and only varied the type of sugar or flour within the crumble recipe to determine which respective type of ingredient led to a better crunch or crisp by running forks across the surface of cooled apple crumble surfaces and recording the resulting sound. After performing Fast Fourier Transforms (FFTs) on the recordings, we observed that plain flour and brown sugar used in the crumble generally produced the crispiest and crunchiest crumble toppings, corresponding to a higher frequency average obtained in the resulting FFT spectrum. However, to achieve a balance in all the desirable qualities of an apple crumble, we would use a mixture of plain and bread sugar

and substitute the brown sugar with white sugar.

2. *For the filling: What methods other than cooking off moisture may we employ to prevent a soggy texture within the filling?* Our group started with only varying the type of apple (green or red) and type of sugar (brown or white) to determine the effect on inherent differences in viscosity in the ingredients on the resulting texture of the filling. The resulting filling was soggy (but brown sugar was soggy) regardless of the type of apple. We, therefore, looked to external thickeners, and results show that the calculated amount of potato starch effectively thickened the filling and removed the soggy texture from the apple crumble filling.

Why this recipe?

Apple crumble has been a longtime specialty of Shiner's Diner, one of the student-run butteries in Yale-NUS College, and one of the most popular specials whenever served. Since one of our group members is part of the buttery and makes apple crumbles weekly, our group wondered how one could improve on the standard recipe used in the buttery, looking for ways to affect the crunch of the crumble as well as the thickness of the filling.

From a culinary perspective, it is important to have a balanced crumble. Crumble that is too dry is extremely flaky without sufficient structural integrity, but focusing too much on drying out the crumble to maximize crunch leaves a rough, hard texture that is unpleasant to eat, and sucks the moisture out of the crumble topping. On the other extreme, the crumble can be too wet, not providing a satisfying crunch and melting into a soppy mess on top of the apple crumble. An ideal balance is maintained by balancing the variables that determine dryness and wetness within the crumble. As for the filling, a thick enough consistency is optimal so as to prevent a soggy texture that is unpleasant, but not too thick such that the

apple crumble becomes a solid block.

In a scientific context, the crunch of the crumble is controlled by the amount of butter and type of flour used. Butter contains a large amount of fat and water, and it is the water that controls the dryness of the crumble. The protein content of the flour across different types also determines the crunch of the crumble, as the proteins in the flour are known to be water-insoluble and instead dissolve in the fat provided by the butter. As for the filling, we can manipulate the viscosity using potato starch. When heated, the potato starch undergoes gelatinization, a process that breaks down the intermolecular bonds of starch molecules in the presence of water and heat, allowing the Hydrogen bonding sites to engage more water. The starch granule irreversibly dissolves, which turns the potato starch into a transparent viscous mixture from its powdery, white appearance, which thickens the apple crumble filling.

The Experiment

Experimental Recipe

We used the following ingredients for our experiment:

✳ 60 apples, 30 green South African, and 30 red Fuji ✳ 6 sticks of butter
✳ 1 bag of white sugar ✳ 1 bag of brown sugar ✳ 1 pack of cinnamon powder ✳ 1 bag of plain flour ✳ 1 bag of top flour ✳ 1 bag of bread flour ✳ 1 bag of potato starch ✳ 1 lemon for juicing

The apple crumble consists of two parts: the filling and the crust. The filling contains cinnamon, vanilla extract, lemon juice, sugar, and apples. The crumble topping consists of butter, flour, sugar, and salt. We did not use rolled oats in the crumble recipe to maximize observation of the effect of flour type on the crunchiness of the crumble. To iterate the experiment

over different variables, you will need several mixing pots for the filling, and cupcake tins to cook the different iterations of apple crumble in.

The recipe for 6 servings is as follows. The recipe makes a relatively small amount, as this was used to fill 3 individual cupcake tins (for the filling) and a single cupcake tin (for the crumble topping).

1. Preheat the oven to 190°C.
2. For the Filling:
 - (a) Peel, core, and slice 3 medium apples. Add 2 tbsps of sugar.
 - (b) Add cinnamon to taste and squeeze a quarter of a lemon. Mix well.
3. For the Crumble:
 - (a) Mix well 50g of flour and 50g of sugar together in a bowl, with a pinch of salt.
 - (b) Slice up 50g of cold, hard butter into small cubes. Add the butter a few cubes at a time into the dry mixture. Using your fingers, incorporate the butter into the dry mixture by breaking the cubes with your fingers and rubbing the dry mixture into the butter.
 - (c) Add more butter cubes and repeat the process until all butter cubes are used, and a shaggy dough is formed.
4. Fill the cupcake tin with the filling and top it off with the crumble topping. Bake for 35 minutes.
5. Let completely cool before serving, or experimenting on!

To this end, our group followed the recipe created above and varied one type of ingredient each time to create an array of apple crumbles with different variables to qualitatively assess which combination of ingredients

provided the crunchiest and least soggy apple crumble (and the best-tasting too!). The created array of apple crumbles is shown in Figure 1.

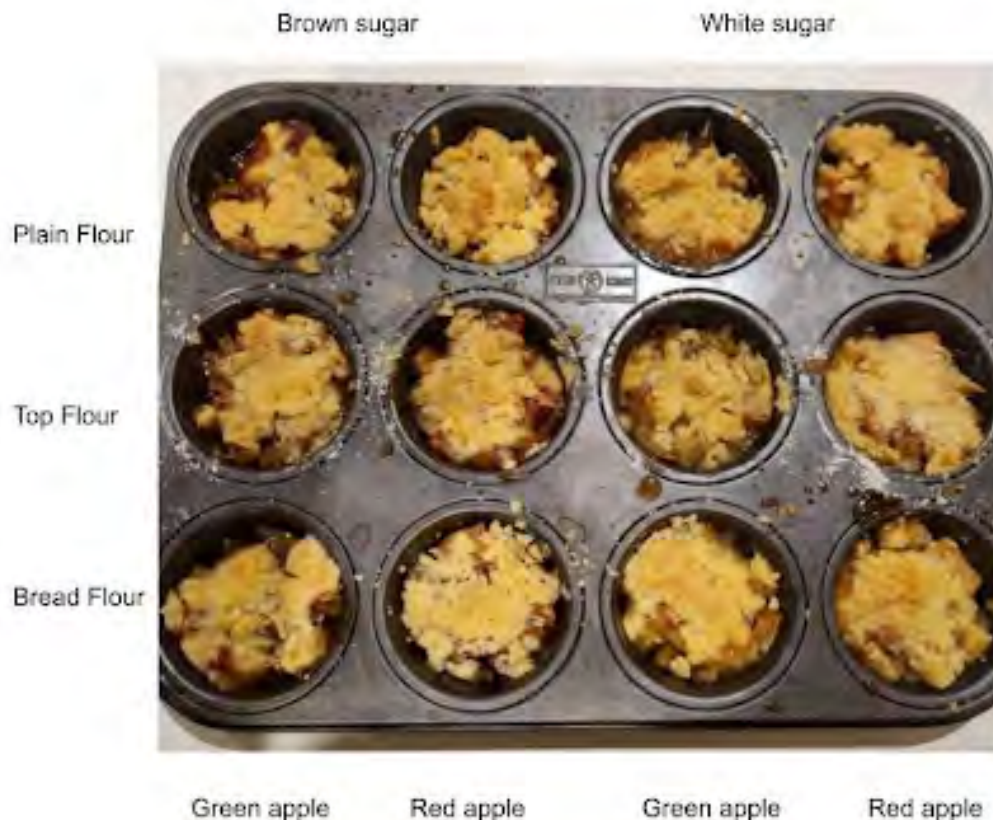


Figure 1: Array of Apple Crumbles made in experimentation stage. Either the type of flour was varied in the crumble, or the type of sugar and apple in the filling. The type of sugar was not varied in the crumble as we knew from the scientific literature that white sugar would lead to a crunchier and crispier texture.

Flour Protein Density

We first test the effect of different flours and their different protein densities on the crunchiness of the crumble. Top flour (5.6g/serving), Plain flour (6.1g/serving), and Bread flour (16.4g/serving) were used for the crumble recipes. Plain flour is our “control” flour as it is the typical type of flour recommended in most apple crumble recipes. Additionally, the protein density of plain flour sits nicely between the Top and Bread flour.

Initial post-baking observations of the array created in Figure 1 were made for the crumble topping made with the different flours. The group agreed that:

Plain Flour

- ✱ Plain flour had the crunchiest crumble.
- ✱ The crumble of plain flour was a tad bit dry and had the driest texture as compared to the other flour varieties.

Top Flour

- ✱ The topping made with top flour can only be described as flaky, without much structural integrity.
- ✱ The crumble fell apart easily and did not have a satisfying crunch when bit into.

Bread Flour

- ✱ The topping made with bread flour was very substantial and had a dense structure, which made biting into very satisfying.
- ✱ The topping was not soggy but was agreed to be the wettest of the three flour varieties.
- ✱ The topping did not make much of a crunch when bitten into.

Qualitative observations made by the group generally supported scientific literature. The proteins in flour are fat soluble, and would thus form a more homogeneous mixture when mixed with the same amount of butter. It follows that bread flour, having the highest protein content, would form the softest crumble topping that was the most 'doughy', and top flour, having the least protein content, would form the crumble topping with a

texture most akin to baked flour (flaky) as there would be fewer proteins that would dissolve in the butter when baked, forming the least homogeneous crumble topping.

Testing Viscosity

From our post-baking observations of the apple crumble array in Figure 1, the group did not observe any difference in the viscosity of filling attributed to the type of apple. Theoretically, the higher pectin content of green apples than red apples would lead to a thicker filling as pectin is a naturally occurring thickener. We decided on green apples for the final apple crumble recipe as the filling tasted better qualitatively.

On the other hand, fillings that used brown sugar had a soggy texture as opposed to those that used white sugar. This is within expectation, as white sugar is generally thicker than brown sugar when made into syrups. We then further explore the difference in viscosity of white and brown sugar in our experiments.

Viscosity Across Sugar Type

In order to measure the viscosity of the sugar mixtures, we measured the time it took for them to slide down a ramp. First, we constructed four different mixtures. Each one contained 100 mL of sugar solution, two with brown sugar and two with white sugar. Then, 15 mg of potato starch was added to each solution. A measuring spoon was then used to take 10 mL of the new solutions and drip them down a plastic ramp, timing how long it took for each solution to reach the end. This process was performed six times for each solution. The results of the experiment for brown sugar and white sugar syrups are shown in the tables:

Brown sugar								
	Time (s)							
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Average	%
Plain	19.05	27.75	24.30	23.50	33.98	29.98	26.43	76.99
Thickener (15 mg potato starch)	34.43	29.5	35.46	25.08	37.16	44.31	34.32	129.88

Length of time taken for brown sugar solutions to fully slide down ramp. The mixture with thickener consistently took longer than the regular mixture in every trial, with about a 30% increase in time over the regular mixture.

White sugar							
	Time (s)						
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Average
Plain	32.01	32.3	44.44	36.71	44.36	37.93	37.96
Thickener (15 mg potato starch)	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Length of time taken for white sugar solutions to fully slide down ramp. The mixture with thickener was too viscous to even fully slide down the ramp, leading to no times being recorded for the white sugar with thickener.

Regarding the viscosity, we found that adding the thickener increased the viscosity of both the white sugar and brown sugar mixtures. For the brown sugar solution, adding the thickener increased the time taken to slide down the ramp by about 4-5 seconds on average, an increase of 30%. For the white sugar solution, adding thickener increased the viscosity to the point where the solution could not fully slide down the ramp. Noticeably, the plain white sugar solution without thickener was more viscous than the brown sugar mixture with thickener; it was about 3 seconds greater on average, an increase of slightly less than 10%.

We then used this data to construct our final apple crumble recipe. First, we had to decide on what type of sugar to use. Through qualitative taste testing, we settled on brown sugar, as we felt it had the right balance of wet and chewy to make a good filling for the crumble. Then, we had to decide on the amount of thickener to add. Again, through qualitative taste testing, we settled on 10 mg of starch per 100 mL sugar mixture. Based on our previous results, 15 mg of starch in 100 mL of sugar provides a 30% increase in viscosity; therefore, 10 mg of starch should provide a 20% increase. Based on our apple crumble recipe, we calculated the volume as follows:

$$V = (6 \text{ apples}) + (\text{water, butter, sugar, etc.}) \approx 1255\text{mL}$$

Given 1255 mL of crumble mixture, and adding 10 mg per 100 mL, we would need to add $1255/10 = 125.5$ mg of potato starch.

Acoustical Analysis of Apple Crumble Crispiness

A secondary facet of our experiment involves perfecting the crumble portion of the apple crumble. Crispiness and crunchiness are two qualities in food that people generally like. Literature suggests that people like crispy foods because they indicate freshness and ripeness in naturally grown foods. Of course, this phenomenon now is manifested through more artificial means like the baking of a crust or the frying of chips.

The crumble is one of the important aspects of an apple crumble. People like to have a crispy surface but brittle enough to crumble in their mouths. We aim to quantify the crumble's crispiness using acoustic techniques known as the fast Fourier transform.

Methodology

1. Bake samples of apple crumble using variations on three different ingredients: Plain, top, or bread flour. Red or green apples. And white or brown sugar
2. Set up a microphone a few centimeters away from one of the sample apple crumble's surfaces and start recording the sound. Make sure your gain is turned up as high as possible
3. Drag a fork across the surface of the apple crumble multiple times.
4. Be sure to record the sounds of the ambient room without scraping any apple crumble.
5. After completing the recordings of all samples, use Audacity or other similar software to run a Fast Fourier Transform on the audio file. This generates an audio spectrum for each crumble sample.
6. Subtract the audio spectrum of the ambient room from the audio spectrum of the apple crumble samples. This gets rid of any noise that may have been generated from air conditioning or other noise.

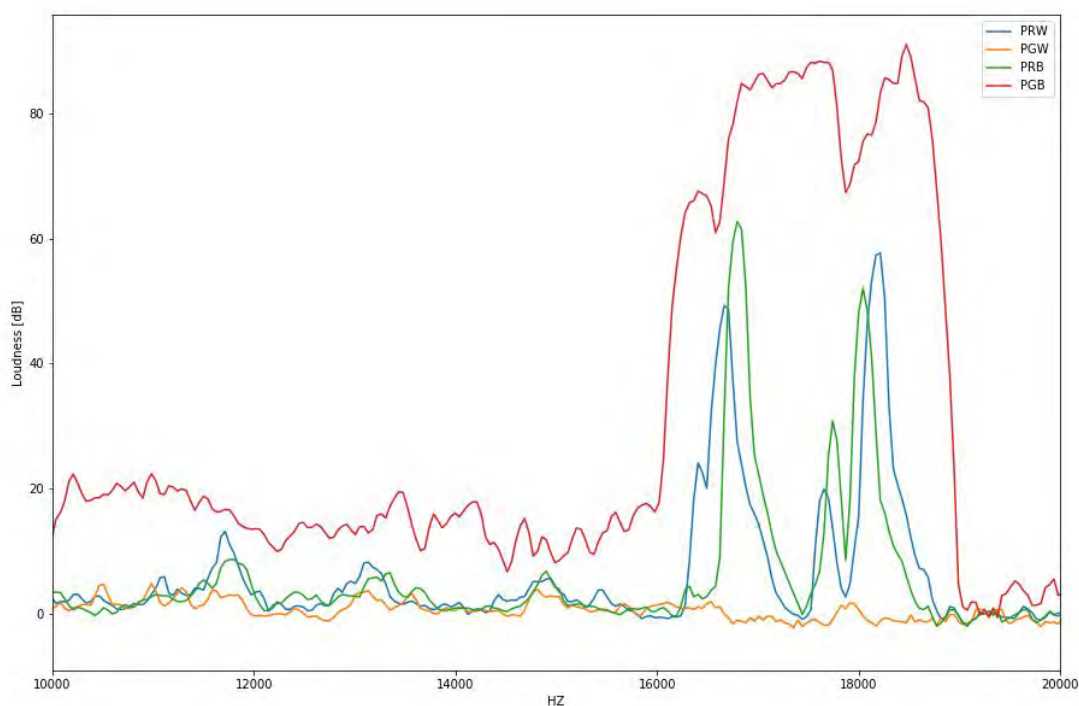
Results

Crispiness is both a psychological and physical phenomenon. When we listened to the scraping of the apple crumble, the team all agreed that plain flour resulted in the crispiest crumble. Psycho-acoustical research in crispiness perception would suggest that this is because those crumbles may have produced higher-frequency sounds at louder volumes.

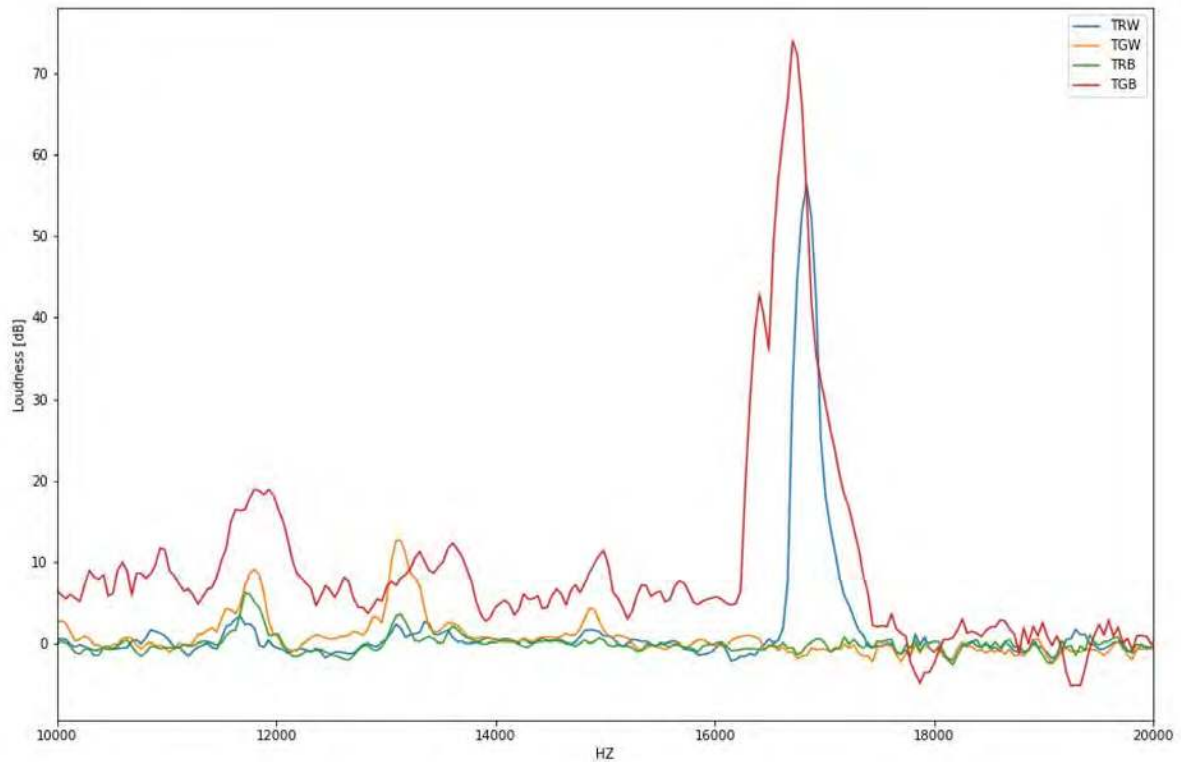
We compare these observations with measurements of the audio spectrum. In the figure, we adopt the following notation in the legend. The first letter would indicate the type of flour used in the sample (plain (P), top (T), bread

(B). The second letter indicates the type of apple (red (R), green (G)). The last letter indicates the type of sugar used (White (W), Brown (B)).

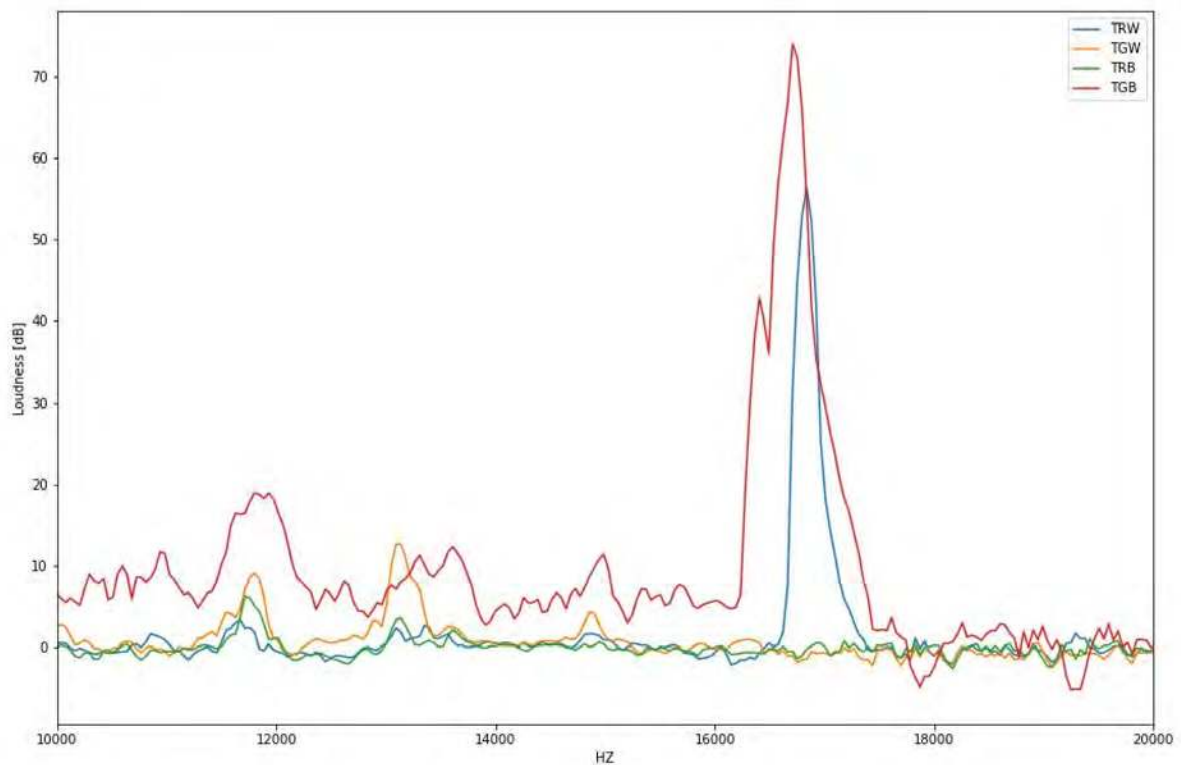
Surprisingly, the graphs do not align with our observations concretely. We had predicted that the plain flour, green apple, and white sugar (PGW) sample would produce the crispiest sound judging from what we heard. However, the Fourier analysis's of this sample revealed no prevalent frequencies where we would expect. Judging from the data, we would estimate that the PGB, BRB, TGB, PRB, and BGW samples produce the loudest and most high-pitched sounds. This may suggest that perhaps brown sugar makes crispier crumble. However, our taste testers seemed to overall prefer the white sugar anyways since the brown sugar sometimes left a strong aftertaste.



Audio spectrum of different apple crumbles with plain flour, as analyzed with a fast Fourier transform.



Audio spectrum of different apple crumbles with plain flour, as analyzed with a fast Fourier transform.



Audio spectrum of different apple crumbles with plain flour, as analyzed with a fast Fourier transform.

Though, there may be many uncertainties not accounted for in this experiment. The speed and force of the scraping of the crumble surface probably vary the sonic profile of the crumble. We did this process by hand, so it is only as constant as our human arms could allow. additionally, there was a lot of noise in the room that could not entirely be removed by the "room" spectrum. We believe this is why some samples did not even produce a sound, or some sounds produced a very broad spectrum. It does seem though in many samples the apple crumble produces two dominant frequencies: one at 17kHz and one at 18kHz.

The important takeaway from this study is from looking at the differences between different flour samples. The different flour samples have different protein contents: Top flour (5.6%), Plain flour (6.4%), and Bread flour (16.4%). We hypothesized that the bread flour would produce the crispest crust because more protein in flour leads to better gluten development. Like bread in bread, we had assumed that this gluten development would help build the structure to the crumble and allow for a crispy surface when the baking process evaporates the starches from the flour. This is not what we found both observationally and qualitatively. The bread flour was the softest crumble out of the samples.

Crispiness is all to do about water control. A crispy object should be devoid of water and prevent water from entering it. We found that plain flour achieved this best. We believe that the protein content must reach a balance of not too much protein or not too little. Enough to develop the gluten to create structure and crispiness, but also not too so that water isn't trapped inside from the strong gluten structure. The open gluten structure is what allows water to escape in many breads with broken-up structures like baguettes and fougasse. We conclude that this is the same in our apple crumble.

While the plain flour had the crunchiest crumble, we actually found the brittle and crumbly quality of the bread flour pleasant as well. That is why in our final iteration of the recipe, we use a mixture of both plain and bread flour, in hopes of having a crispy surface of the crumble, but a substantial and crumbly structure.

The Final Apple Crumble Recipe

With all of the findings of our results, we construct the final apple crumble recipe while being informed of the following findings and considerations:

- ✱ The final recipe uses a ratio of 1:1:0.8 of sugar, flour, and butter.
- ✱ Results of the FFT graphs guide us to use plain and bread flour in our final recipe.
- ✱ A ratio of 75% plain flour and 25% bread flour provides us with the crispy and crunchy texture in the crumble provided by the plain flour but also the structural integrity and satisfying mouth feel provided by the bread flour.
- ✱ Qualitative reasoning guides us to choose green apples with brown sugar in the filling as they tasted the best out of the array of apple crumbles.
- ✱ The increased sogginess provided by the brown sugar is counteracted by the calculated potato starch amount, which thickens the mixture sufficiently.

Ingredients

Filling

✳ 6 green apples, cored and sliced ✳ 70g brown sugar ✳ Cinnamon powder, to taste ✳ 125.5 mg of potato starch ✳ Juice from half a lemon

Crumble Topping

✳ 113g cold cubed butter ✳ 112g plain flour ✳ 38g bread flour ✳ 150g white sugar ✳ A pinch of salt ✳ A pinch of cinnamon

Directions

1. Preheat the oven to 190.

2. For the Filling:

(a) Peel, core, and slice 6 medium green apples.

(b) Add the rest of the ingredients and mix well.

(c) Sieve the potato starch into the mixture and mix gradually until well incorporated.

3. For the Crumble:

(a) Mix well both flour and sugar together in a bowl, with a pinch of salt.

(b) Slice up 113g of cold, hard butter into small cubes. Add the butter a few cubes at a time into the dry mixture. Using your fingers, incorporate the butter into the dry mixture by breaking the cubes with your fingers and rubbing the dry mixture into the butter.

(c) Add more butter cubes and repeat the process until all butter cubes

are used, and a shaggy dough is formed.

4. Fill the baking tin with the filling and top it off with the crumble topping. Bake for 35 minutes.
5. Serve right out of the oven with a scoop of ice cream! Yummy!



The optimized apple crumble!

Conclusion

The takeaways here lead to the final product, being composed of distinct quantities of each ingredient to ensure maximum crispiness. A specific ratio of 1:1:0.8 of flour to sugar to butter was used to create this product. The flour was a combination of 25 percent bread flour combined with 75 percent plain flour. Only white sugar was used for the crumble and green apples for the filling. According to FFT data, plain and bread flour contributed roughly the same in regards to the crispiness of the crumble. However, accounting for individual preference, the type of flour contributed also to the internal structural integrity of the crumble. A combination of plain and bread flour combines the strong structural integrity of plain flour with some softness from the bread flour. This optimizes and finds a balance between crispiness and also a softness to the crumble. Previous studies conducted found support for brown sugar's lack of viscosity in comparison to white sugar, which supported the use of potato starch in increasing the viscosity of the filling. Potato starch was added to remove sogginess and was effective in doing as such without altering the taste of the filling and only affecting the viscosity. Green apples were preferred over red due to their lack of sweetness. Molecularly, green apples have a greater protein content than red apples, along with a thicker skin which attributes to the comparatively higher tartness of green apples. As people tend to stray away from overwhelmingly sweet foods, a combination of brown sugar to minimize sogginess and green apple for adequate sweetness, along with potato starch to thicken the otherwise soggy filling provided us with the perfect combination.

All in all, these modifications were successful in resolving the presented dilemmas. New substitutes could be introduced in place of potato starch, but there was insufficient time to explore this possibility. Other apple types could also be used, as there exist over 80 types of red apples and tens of

green apples. The structural makeup of apple types determines both the water content and sweetness profile. A future experiment may look into categorizing and examining apple types, which may create the “perfect” apple crumble. But overall, this experiment proved successful in dissecting the recipe and modification process, in order to create a delicious product.

Aquafaba Meringues



By Arjav and Ying Yi



Preparation time 20 minutes



Baking time 75 - 90 minutes



Total time 1.5 - 2 hours



Serves 10

Summary

Classic meringues are delicious but very fragile, and problems often occur with collapsing egg foams. This is even more so a problem with egg substitutes such as aquafaba.

We studied the effects of different types of stabilizer on whisked aquafaba to determine how best to adapt recipes to be vegan-friendly.

Why this project

Egg whites are an essential structural component of many dishes and desserts. However, this is often a problem for vegans who do not consume egg whites.

While there exist egg substitutes like aquafaba, they tend to be even more delicate and unstable than egg whites. Additionally, they have been studied much less than egg whites. Thus, we wanted to investigate the effects of different stabilizers that are traditionally used on egg whites on aquafaba.

We wanted to find out what stabilizer or combination of stabilizers would be best for stabilizing aquafaba foams for different purposes.

Our experiment

We obtained aquafaba from canned chickpeas and separated it into four equal-sized samples.

One sample was set up as the control. To the other three, we added sugar, cream of tartar, and lemon juice, respectively.

We then whisked the aquafaba to stiff peaks and decanted it into cups, after which we observed it as it sat and collapsed over time.

We also observed the aquafaba foam structure under the microscope.

Our discovery

In the span of 20 minutes, the control, which had no additions, had collapsed to half its initial volume. The other three, however, showed little noticeable change in volume.

We noticed that the added sugar produced a denser foam. This aligns with what we would theoretically expect, with sugar making the aquafaba more viscous, making it harder for the bubbles to merge and escape, resulting in a denser foam. This makes sugar an excellent additive to stabilize deserts that are expected to be sweet and can benefit from the increased structure that a denser foam would provide.

The cream of tartar produced a stable foam over the time observed (about 1 hour) and was the least intrusive additive, neither contributing to a sweet taste nor an acidic flavor. It also only required a very small amount to be added. Thus, it is suitable for dishes that should be neither sweet nor acidic.

The lemon juice also produced similar results in terms of stability (at least for the observed time period). However, it imparted an incredibly sour, unpleasant flavor to the meringue that would be rather difficult to cover up if not desired.

The acids (cream of tartar and lemon juice) work by decreasing the pH of the albumins (proteins present in aquafaba that allow for foaming). This allows them to denature easier and therefore foam faster. Additionally, the added H^+ prevents the protein molecules from binding too tightly. This prevents the water from being squeezed out of the foam and the trapped air bubbles from escaping.

Summary and lingering questions

To stabilize aquafaba foams, the best choice of stabilizer would be determined by first by seeing if the dish already requires sugar or lemon, and using the cream of tartar as an all-purpose stabilizer and can be used additionally for better results. We decided that using sugar and cream of tartar together would be ideal for making meringues.

We would also like to investigate the effect of different quantities of the additives and the effect of using them in conjunction with one another.

Ingredients

✳ 400g chickpea water (i.e., two cans of chickpeas) ✳ 1 teaspoon cream of tartar ✳ 100g sugar ✳ 1 teaspoon vanilla extract

Equipment

✳ Metal Mixing Bowl ✳ Strainer ✳ Whisk/Stand Mixer ✳ Oven ✳ Baking Tray ✳ Parchment Paper ✳ Spoon/piping back

Directions

1. Preheat the oven to 110°C/90°C.
2. Drain the cans of chickpeas over the mixing bowl, using the strainer. Chickpeas can be saved for a future meal.
3. Add the cream of tartar to the aquafaba (chickpea water).
4. Using either the stand mixer or the whisk, mix the aquafaba till soft peaks begin to form.
5. Add the sugar in small increments while whisking continues.
6. Once stiff, glossy peaks begin to form, add the vanilla form.
7. Stop whisking when all the foam has reached the stiff peaks consistency. One good, but risky, test for this is to hold the bowl upside down over your head.
8. Line the baking tray with a piece of parchment paper.
9. Use a spoon or a piping bag to pipe the mixture onto the baking tray in evenly sized blobs. Flourishes are recommended
10. Bake the vegan meringue for 75 minutes until it is crisp.
11. Serve the meringue with anything (including potentially the chickpeas saved earlier) on the side.

Pesto Sauce



By Alvin, John, Zen



Prep time 5 minutes



Cook time 5 minutes



Serves 3–4

Summary

Pesto pasta is one of our favorite pasta, with a flavor profile and texture complexity that puts Carbonara and Aglio Olio to shame. However, pesto pasta contains a high amount of Calories due to the amount of oil required to make the pesto sauce. The pesto sauce is also an unstable emulsion that leads to the oil separating from the sauce after only a short period of time. We experimented with ways to maintain the taste of the sauce while reducing oil usage. We also opted to use ingredients that have emulsifying properties to stabilize the emulsion. We find that egg yolk and xanthan gum are good options that can create stronger emulsification in the pesto sauce. Additionally, egg yolk adds a desirable, creamy mouth-feel to the pesto pasta, while containing fewer Calories than oil. We conclude that adding egg yolk to the pesto sauce enhances the recipe while making it healthier.

Why this project

Pesto pasta is one of our favorite pasta, with a flavor profile and texture complexity that puts Carbonara and Aglio Olio to shame. However, the

amount of oil used to make a portion of pesto sauce is astoundingly high! As such, we wanted to create our own recipe of pesto sauce that reduced the amount of oil, and by relation Calories, while still ensuring that the taste was not compromised. At the same time, we noticed that for people that take longer to finish a meal, the oil from the pesto sauce might separate before the food is completed if the emulsification is not stable enough. This means that the enjoyment of the sauce might not be consistent throughout the meal, and we aimed to stabilize the emulsion by adding emulsifiers. Hence, we sought to address both the high Calories and weak emulsion together. We experimented with egg yolk and xanthan gum in two different recipes.

Our experiment

We bought sufficient ingredients from the supermarket to make 3 different pesto sauces, with each portion containing 4 servings worth of pesto sauce. Our first pesto sauce was our control, which we made by following a recipe online. We then made our second and third sauces, replacing $\frac{1}{2}$ of the oil used with 2 large egg yolks for the second sauce, 2 grams of xanthan gum, and $\frac{1}{4}$ cup of water for the third sauce.

We assess our results both quantitatively and qualitatively. For our quantitative analysis, we calculate the total Calories reduced for the second and third sauces compared to the first one. For qualitative analysis, we then compared the different sauces for 3 main categories, namely color, texture, and taste.

Our discovery

Calculating the Calories in each sauce, we found that compared to the first sauce that follows the original recipe for pesto sauce, the second and third sauces had 368 kcal and 477 kcal lesser, respectively. This is due to the

amount of oil being reduced, which made up almost 70% of the Calories in the first sauce. In the second and third sauces, the olive oil made up about 45% and 50% of the total Calories, respectively. This reduction in Calories is countered by 110 kcal from the two eggs in the second sauce, while the xanthan gum in the 3rd sauce is almost negligible (< 1 kcal).

As for our qualitative results, we found that in the first sauce, the oil did indeed separate from the pesto sauce after about 20 minutes. We also note that the color is a rather dark green sauce and is very fluid.



Sauce 1 (original recipe, left), Sauce 2 (with egg yolk, middle), Sauce 3 (with xanthan gum, right)

For the second and third sauces, the viscosity of both was much higher, causing the sauce to appear similar to a paste than a simple liquid. Both sauces were also of a lighter color than the first sauce.

However, when mixed into linguine pasta, we noticed that the pasta with sauce 2 had a desirably creamy texture while the pasta with sauce 3 was relatively dry. We also note that the layer of oil observed on the top of sauce 1 did not form in sauces 2 and 3.

Under the hood

In the first sauce, water (mainly from the basil leaves and crushed garlic) and oil (mainly from the olive oil, crushed garlic, and pine nuts) form a partial water-in-oil emulsion. This is formed by lipid molecules from the basil leaves and fat crystals from the olive oil that has emulsifying properties. These molecules have hydrophilic and hydrophobic ends that are attracted to oil and water respectively, separating the two. This leads to emulsification, with droplets of water suspended in oil.

However, after some time, the emulsion breaks as it is not stable. Egg yolk and xanthan gum increase the emulsification activity as the former contains proteins such as lecithin while the latter is a polysaccharide. Both these proteins and polysaccharides have hydrophilic and hydrophobic properties, thus stabilizing the emulsion.

Conclusion

As discussed above, both egg yolk and xanthan gum have emulsifying properties that can stabilize the emulsion in pesto sauce. This prevents the oil from separating from the sauce mid-meal and ensures a consistent flavor throughout the dishes where the pesto is mixed. Additionally, the second sauce has a creamier mouthfeel that is desirable in pesto pasta, making the second sauce the best-tasting one. An interesting point to note was that the third sauce, which used xanthan gum, was a little too dry, even though we added $\frac{1}{4}$ cup of water into the mix. That speaks to the strength of xanthan gum as an emulsifier but does make the pesto pasta slightly less tasty. A potential solution to this would be to increase the number of basil leaves used to provide more water for the sauce but not risk diluting the taste of the sauce too much by just adding water alone.

Furthermore, the total reduction in calories in sauces 2 and 3 are 368 kcal

and 477 kcal, respectively, for a batch of sauce that serves four. This results in an even lesser reduction in calories per meal. Ultimately, even though the third sauce reduces more calories, this reduction is not a sizeable one. Personally, the trade-off for a few calories to have a creamier, tastier pesto pasta is absolutely worth it.

Ingredients

✳ 2 cups fresh basil leaves ✳ $\frac{1}{2}$ cup Parmesan cheese ✳ $\frac{1}{4}$ cup olive oil ✳ 2 large egg yolks (pasteurized) ✳ $\frac{1}{3}$ cup pine nuts & walnuts (An alternative is to use almonds instead of pine nuts; cheaper option with just as good a taste!) ✳ 5 cloves of minced garlic ✳ Salt and pepper

Directions:



Healthy pesto pasta!

1. Blend the fresh basil leaves with the pine nuts/almonds and pulse them several times while pushing down the mix stuck to the slides of the blender
2. Add in the minced garlic and the Parmesan cheese and pulse more while continuing to scrape the sides of the blender
3. Pour in your olive oil intermittently while continuing to pulse the mix
4. Add in your egg yolks and blend for 30 seconds before scraping the sides and pulsing a few more times
5. Season with salt and pepper to taste. You now have a healthier yet creamier pesto sauce! Toss with pasta and add your protein of choice to get your own creamy pesto pasta.

Mala Xiang Guo



By Dylan, Meryl, and Pache



Prep time 15 minutes



Cook time 15 minutes



Serves 2

Summary

Mala Xiang Guo is a highly popular dish in Singapore. Its addictive quality comes from the spicy kick and numbing sensation one experiences when eating Mala. Nevertheless, as a relatively new dish, there is currently no consensus on what constitutes ‘good’ Mala. With that, we conducted an experiment to fulfill the following aims and research questions:

1. To investigate in greater detail what causes the numbing sensation (má) when eating Mala.
2. To conduct an experiment to discover how consumers’ perceptions of the taste and mouthfeel of Mala change as the concentration of peppercorns increases.

We hope our experiment can generate insights with respect to consumers’ preferences for Mala sauce. Overall, our analysis suggested that while adding more Sichuan peppercorn may increase the perceived tastiness of Mala, this effect is only effective up to a certain degree: adding too much peppercorn will cause the taste of Mala to become slightly unpleasant.



Mala Xiang Guo

Why this project

Mala is a favorite spicy food amongst Singaporeans: the fiery kick and numbing sensation one gets from eating Mala is thought to be what gives Mala its addictive character.

The substances that cause the spicy and numbing sensations in mala are capsaicin in the red chili (for spice) and alpha-hydroxy-sanshool in the Sichuan peppercorns (for numbing). These compounds bind to a unique set of nerve receptors on our lips, allowing for the distinct spicy and numbing sensations experienced while eating Mala.

Upon contact, these nerve receptors send electrical currents to our brains, tricking us into believing we are experiencing sensations like burning and vibrating lips. Capsaicin binds to tactile receptors known as VR1 Receptors, whereas Hydroxy-alpha-sanshool binds to Meissner receptors, both found in our lips. Endorphins are also released as a result of the stimuli, making the experience of eating Mala one that is highly exhilarating and

pleasant.

For our experiment in specific, we focused on how the amount of Sichuan peppercorns added (producing the numbing sensation) would affect the other sensory and taste experiences of the Mala.



Our Sichuan Peppercorns!

Our experiment

Ingredients

✧ 200ml Vegetable Oil ✧ 1/8 cup of peppercorn ✧ Cinnamon ✧
Ground Chilli ✧ Ground Sichuan peppercorn ✧ Ground Coriander ✧
Ground ✧ White Pepper ✧ Chopped Garlic ✧ Star Anise ✧ Beancurd
Skin ✧ Instant Noodles ✧ Vegetables ✧ Coriander



The Mala We Cooked!

Procedures

1. We first divided the stir-fry ingredients (instant noodles, beancurd skin,

vegetables) into three equal portions for the three samples we plan to make. Sample A would be our control, containing zero peppercorns (zero Má). Sample B would contain a small number of peppercorns (Small Má). Lastly, Sample C would contain the most amount of peppercorns (Big Má) Sample C: Big má (a lot of peppercorns). To achieve this, we divided 80g of peppercorns into a ratio of 1:3, where the smaller amount (20g) went into Sample B, and the larger portion went into Sample C (60g).

2. After preparing the ingredients, we had to fry out the Mala base sauce. The miscellaneous ingredients in the sauce (oils, salt, chili powder, cinnamon, coriander, white pepper, chopped garlic, salt) were kept constant across the three samples of mala sauce; only the quantity of peppercorns used varied as outlined above. The time to fry the sauce was also kept constant (5 minutes).

3. Following the creation of the three base sauces, we then proceeded to stir fry them with instant noodles, vegetables, and beancurd skin to make them more palatable for our consumer sample. We stir-fried every sample for the same amount of time (10 minutes), producing the three samples:

✧ Mala A (Control: No Sichuan peppercorns added)

✧ Mala B (20g of ground Sichuan peppercorns)

✧ Mala C (60g of ground Sichuan peppercorns).

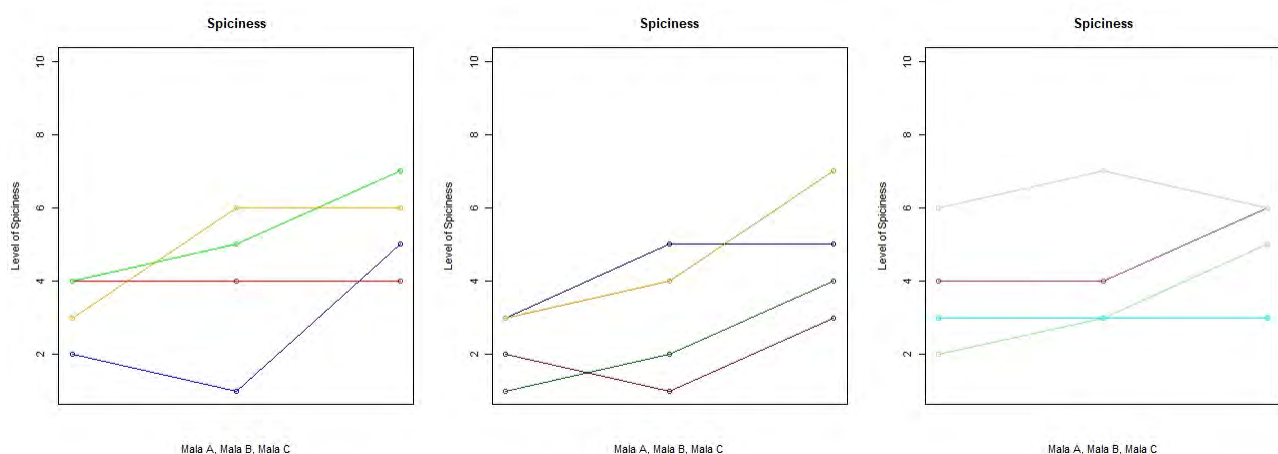
4. *Tasting:* To select our sample of consumers, we conducted convenience sampling. We first invited passers-by in the Saga Buttery to try our dish, then invited friends and strangers around the College to partake in the tasting session. It was a within-subjects design in that every consumer tasted all three samples. To minimize demand characteris-

tics and social desirability biases, consumers were not informed our experimental aims. The order in which the samples were served to consumers was also counterbalanced.

As consumers tasted the Mala samples, they concurrently rated the samples on a Google Survey form. The form asked them to rate the level of (1) spiciness, (2) numbness, and (3) tastiness of every sample on a 10-point Likert scale, where 1 = the least spicy/numb/tasty, and 10 = the spiciest/numb/tasty. To minimize any transfer in-between samples, we requested consumers to drink a cup of water in-between sampling sessions.

Our discovery

Overall, our analytical sample comprised 12 Yale-NUS College students who visited the Saga Buttery to try our Mala. Of the three Mala samples, there was a general perception that the Mala with the most Sichuan peppercorns was spicier than the rest of the samples, even though there was no change in the total amount of chilli peppers applied (we only varied the amount of Sichuan peppercorn)! Additionally, there was a significant variation in perceived tastiness of the Mala as the concentration of Sichuan peppercorns increased.



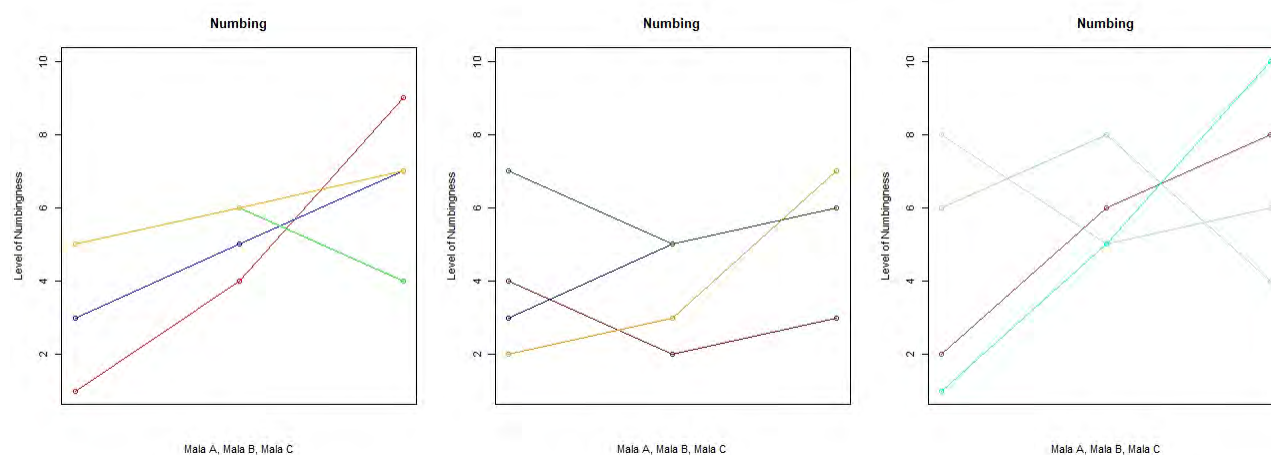
Spiciness Ratings across Three Malas

Consumer Perceptions of Spiciness

Surprisingly, we found that **most of the tasters found Mala Sample C** (the Mala with the highest concentration of Sichuan peppercorns) **to be much spicier** than Mala Samples A and B which had no Sichuan peppercorn and less Sichuan peppercorns respectively. This finding suprising given the fact that the total concentration of chilli peppers across all three Mala samples remained constant.

Consumer Perceptions of the Numbing Effect

Interestingly, some consumers felt that the control Mala (Mala A) was more numbing than Mala B. We posited that there might have been a placebo effect induced by the chili pepper's spiciness that caused people to experience the numbing sensation of Mala despite the absence of any chili peppers.

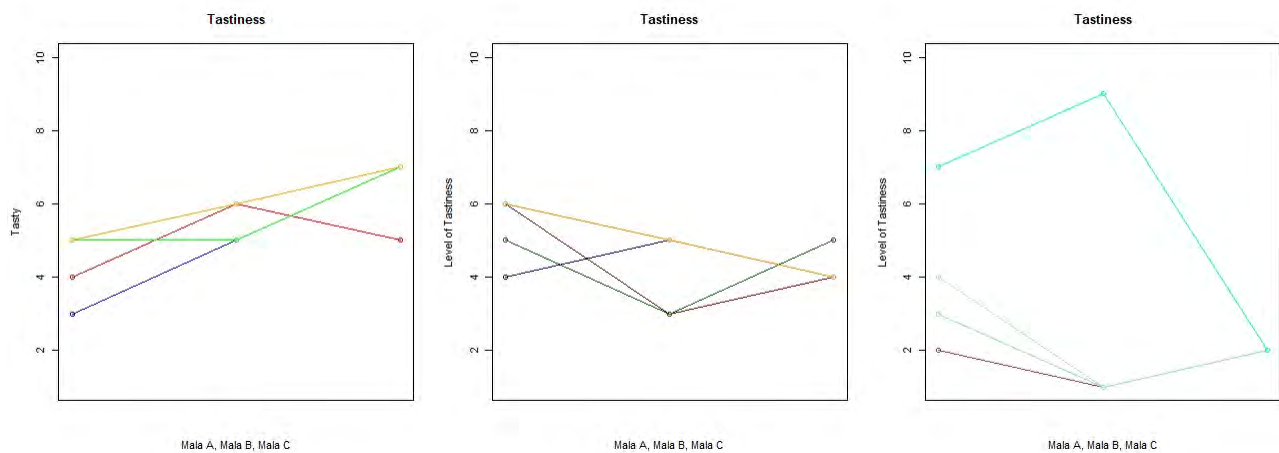


Numbing Effect Ratings across 3 Malas

Consumer Perceptions of Tastiness

We found a significant amount of variation among tasters on the perceived tastiness of the Mala as we increased the concentration of peppercorns. In general (even among those who liked a stronger taste of Mala), consumers found the tastiness of the Mala sample with the highest concentration of

peppercorns less pleasant relative to the samples with less or no peppercorns.



Tastiness Ratings across 3 Malas

Our Reflections

Overall, we found highly mixed reviews on the tastiness of the Mala samples as the concentration of Sichuan peppercorns increased.

Why Did The Mala Get more *Spicy*?

We postulated that there could have been a synergistic effect between the hydroxy-alpha-sanshool in the Sichuan Peppercorns and capsaicin in Chilli Peppers that led to consumers perceiving Sample C (most peppercorns) to be the spiciest.

As the concentration of Sichuan peppercorns increased, the numbing of Meissner receptors further intensified. This might have intensified tasters' perceptions of the spicy effect, causing them to perceive the Mala became spicier!

The Variation in Tastiness

From our findings, we found that *there is no 'optimal' level of numbing-ness (má)* that consumers find most preferable for their Mala.

Too much Sichuan peppercorns seems to result in an over-stimulation of nerve receptors, resulting in taste becoming less pleasant. What was most likely was that consumers' subjective tolerances and preferences of numbness and spiciness might've contributed to the mixed findings on tastiness. For example, while some consumers may have found a high level of numbness exciting and delicious, someone with a lower tolerance for numbness might've found that less palatable.

Limitations of Our Experiment

Our findings are qualified by several limitations. Firstly, we did not take into account the temperature at which the samples were served. As the tasting session occurred in an air-conditioned environment (Saga Buttery), the three samples progressively got colder as time passed on. Hence, consumers who arrived later might've perceived the samples as less 'heaty'. Future research should consider ways to keep the temperature of the three samples consistent as time passed, such as using heating lamps to keep the food warm.

Importantly, we had a tight budget and hence were limited in the amount of ingredients we could buy. Hence, we could not create larger servings of Mala, contributing to a relatively small sample size of 12 consumers. Furthermore, we conducted convenience sampling instead of random sampling, which might have introduced a sampling bias into our experiment. These factors thus limit the generalisability of our findings to the larger population. Future research should thus conduct more systematic and large-scale recruitment of consumers, as well as to try to achieve a more representative sample in terms of race, gender, and age.

Lastly, we tried to minimize the lingering numbness as consumers moved from one sample to another by getting them to drink water. Moreover, we counterbalanced the order by which the samples were served to consumers.

Nevertheless, future research can explore ways to minimize the lingering numbness even further, such as providing milk for consumers to drink, as well as making consumers wait for a longer amount of time (e.g., 15 minutes) before moving onto the next sample.

Marinating Beef Steak



By Anthea and Brian



Preparation time 1-2 hours



Serves 4–6 (3 ribeye steaks)

Summary

We sought to find a more budget-friendly way to make cheap, tough cuts of beef steak tender and delicious using science! To do so, we tested the effect of different acidity levels of marinades and time on meat tenderisation. We found that using acid as a marinade for steak is generally an effective way to tenderize meat, but other factors, such as natural enzymes found in certain fruits, can improve the tenderisation. We also discovered that the amount of time the meat is left in acidic marinades affects the tenderisation — too long, and the meat can start becoming tough again. We conclude that with the right marinades, college students can turn frozen low-price steaks into restaurant-quality steaks.

Why this project

Any meat aficionado will tell you that the most important thing for a good steak is choosing the proper cut of meat. However, the ‘right’ cuts can be extremely expensive, to the extent that cost is prohibitive for us college students.

We, therefore, set out to find the best way to tenderize the meat while enhancing its flavor, to make a cheap cut of steak taste ‘expensive’. During

our preliminary research, we considered other methods, such as physically pounding or aging the meat, but eventually settled on the most low-effort, low-cost, and time-saving procedure — finding the right marinade. We were curious about how different acidity levels and time would affect the breaking down of the meat.

Our experiment

We used 3-dollar frozen ribeye steaks that we cut up and soaked in 4 different marinades of varying levels of acidity: olive oil (pH 5), red wine (pH 4), pineapple juice (pH 3), and lemon juice (pH 2). A control setup with no marinade was also used to test the original toughness of the meat. We then tested the tenderness of the meats at the 3-hour and 6-hour mark of soaking in the marinades to compare the effects of both time and acidity. We used matrices easily replicable at home to measure tenderness, such as the ‘5-finger test’. We attributed a toughness scale between 1-5 for each finger so that home cooks can replicate the same test with different recipes (See Figure 1). We also tore the meat samples apart with a knife and fork to see how easily they could pull apart.

Our discovery

Our preliminary hypothesis that the more acidic the marinade, the more tender the meat, was proven to be generally correct. At the 3-hour mark, the control steak was the toughest, followed by the olive oil, red wine, lemon juice, and pineapple juice (See Figure 2). This is because acid is able to denature and unwind protein strands in the meat, breaking it down and making it tender. The broken-down protein strands can also trap more liquid between them, making the meat juicier and absorbing the flavor of the marinade (See Figure 3). However, we had two surprising findings.

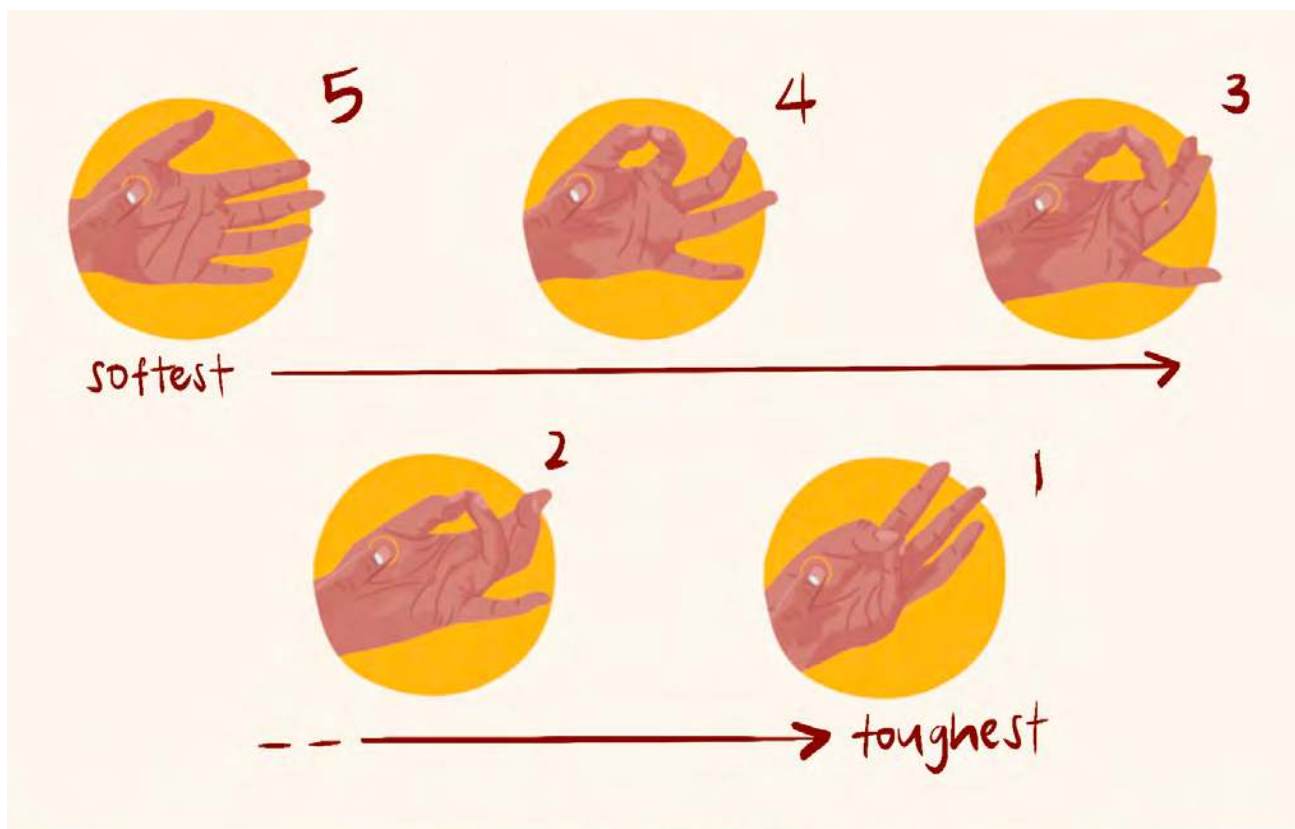


Figure 1: Finger test for ranking tenderness levels

First, even though lemon juice was the most acidic, it was not the most effective — instead, pineapple juice was the most effective. We discovered this was due to the enzyme ‘bromelain’ occurring naturally in pineapple, which contains proteases that digest the protein in meat by breaking up peptide bonds (See Figure 2). From our research, we also learned that other foods contain proteases that could be a substitute for pineapple. For example, papaya contains papain, which is another group of proteases, and miso and honey both contain digestive enzymes that can break down protein.

Second, at the 6-hour mark, the steaks in the lemon and pineapple juice marinades were tougher than at the 3-hour mark, while the ones in olive oil and red wine became more tender. We learned that the tenderizing effect of acidic marinades is not linear over time and can actually reverse if the meat is marinated for too long. This is because the high acidity over a long period of time causes the protein bonds to re-tighten and expel liquid, making the final result chewier and drier.

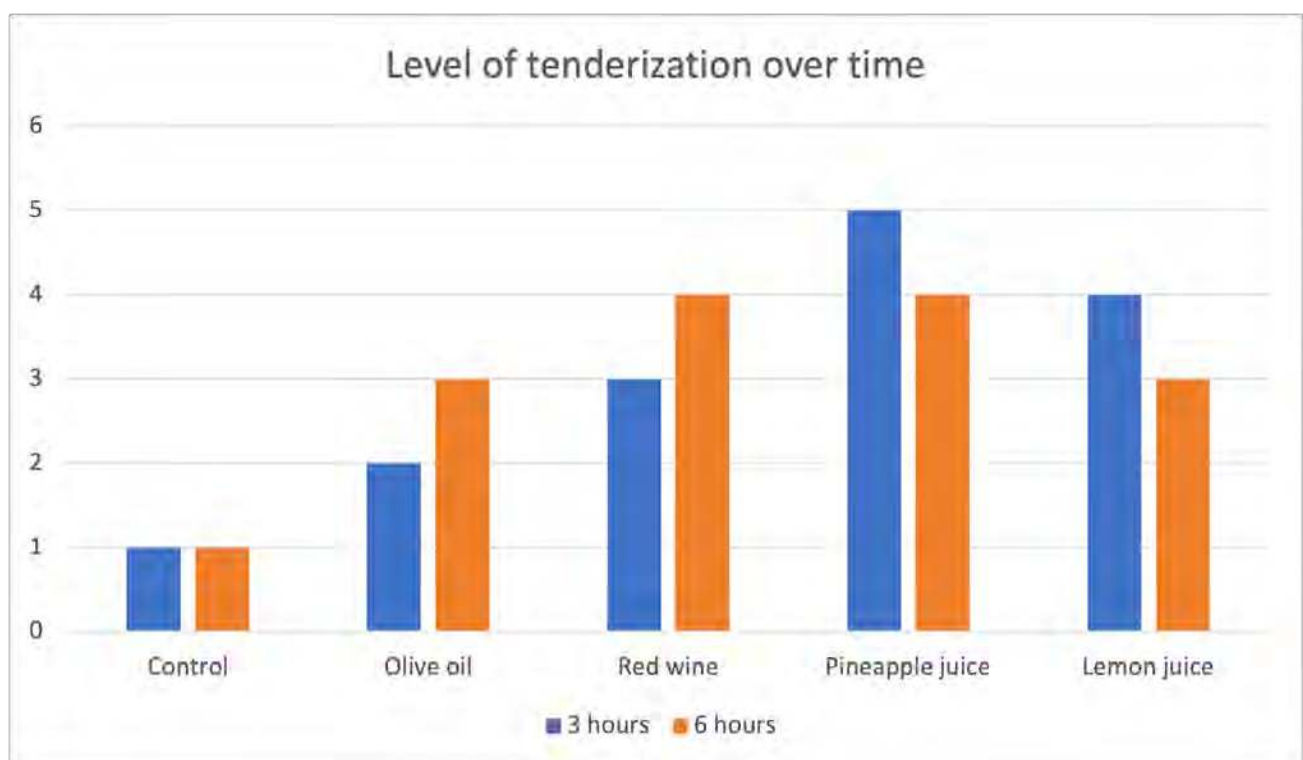


Figure 2: Level of tenderization over time measured using the 5-finger test and ranking

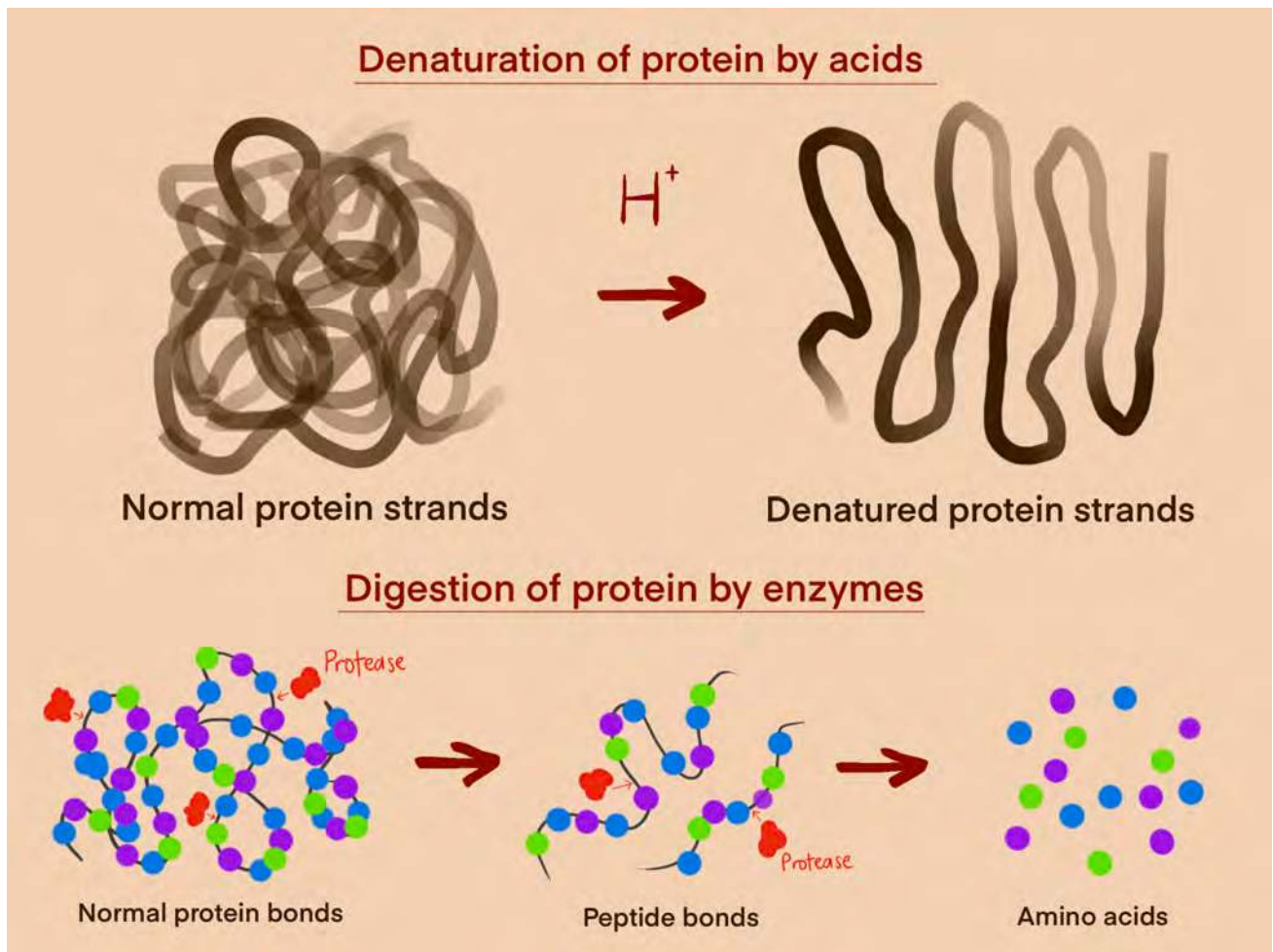


Figure 3: Effect of acids and enzymes on protein

Conclusion and lingering questions

We found that there is no single optimal acidity as all the marinades had some tenderizing effect – the best marinade for you will depend on how much time you have. If you prefer to leave your meat overnight, less acidic marinades (e.g. olive oil) can tenderize it over 6-8 hours. If you are pressed for time, highly acidic marinades (e.g. lemon juice) will do the job in 2-3 hours. If you are really short of time, acidic marinades containing enzymes (e.g. pineapple) can tenderize the meat in 1-2 hours since the combined effect of high acidity and natural enzymes are the strongest. To expand on this study in the future, maybe we can also consider the effect of alkaline marinades (e.g. sodium bicarbonate) on meat to see if there is a similar tenderizing effect. Alkalies are known to increase the rate of the

Maillard reaction that occurs between amino acids and sugars which gives the browning effect when cooking, and it may be interesting to find out if this will help in getting a good sear on a steak.

Ingredients

✳ 3 ribeye steaks (cheap ones) ✳ 150ml pineapple juice ✳ 100ml olive oil (optional, but will make meat moister) ✳ 1.5 teaspoons salt ✳ 1.5 teaspoon ground black pepper ✳ 2 cloves garlic, minced ✳ 3 teaspoons of your desired herbs/spices (eg. paprika, oregano, onion powder) ✳ Cooking oil (for frying)

Equipment

✳ Large ziplock bag ✳ Kitchen towels ✳ Pan and stove

Directions

1. Add all the ingredients for the marinade into the large ziplock bag.
2. Add the steaks into the ziplock bag, and squeeze as much excess air out of the bag as possible so that all the meat is soaking in the marinade. Leave the ziplock bag in the fridge for 1-2 hours.
3. After the meat has been soaking for 1-2 hours, remove the steaks from the ziplock bag and pat the steaks dry with kitchen towels. Sprinkle some salt and pepper on the steaks.
4. Add cooking oil to a pan and preheat it using high heat. Once the oil 'shimmers', add the steaks to the pan. You should hear a loud sizzling.
5. For a medium rare steak, cook the steak for about 2 minutes on each side.
6. Once cooked, remove the steak from the pan and allow the meat to

rest for 10 to 20 minutes.

7. After resting, slice the steak against the grain and serve!

Butter Chicken



By Ozair, Yi Jia and Jay



Prep time 120 minutes, excluding marination time



Serves 6–8

Summary

Butter chicken stands out as an iconic dish originating from South Asia that continues to make the mouth water in restaurants worldwide. In a traditional recipe of the dish, cashew nuts are integrated to add a creamy thickness and a rich flavor. In our investigation, we sought to determine how changing the form of cashew nuts used in a butter chicken recipe could affect the thickness of the overall gravy. In doing so, we cooked butter chicken with three variations of cashew nuts (whole, cut, and ground) and cornstarch to determine the impact of cashew nuts on the thickness and taste of the sauce. We found that ground cashew nuts resulted in the creamiest and tastiest butter chicken sauce, and hence, we incorporated this in our suggested recipe.

Why this project

In a cosmopolitan city such as Singapore, the butter chicken serves as a gateway to the taste profiles of many South Asian cuisines, and it is for this reason that we were keen to discover what recipe could be used to potentially escalate the taste further while also attempting to bring it closer to its cultural roots. Cashew nuts are often claimed to be responsible for

the thickness of a butter chicken recipe; we wished to put this to the test by measuring how different forms of cashews would affect the sauce's thickness. We learned during the milkshake experiment (in Lab 6 of YSC 1223) how thickeners could be added to food to change its texture. We thus wished to apply that to find out how to make butter chicken thicker.

Our experiment

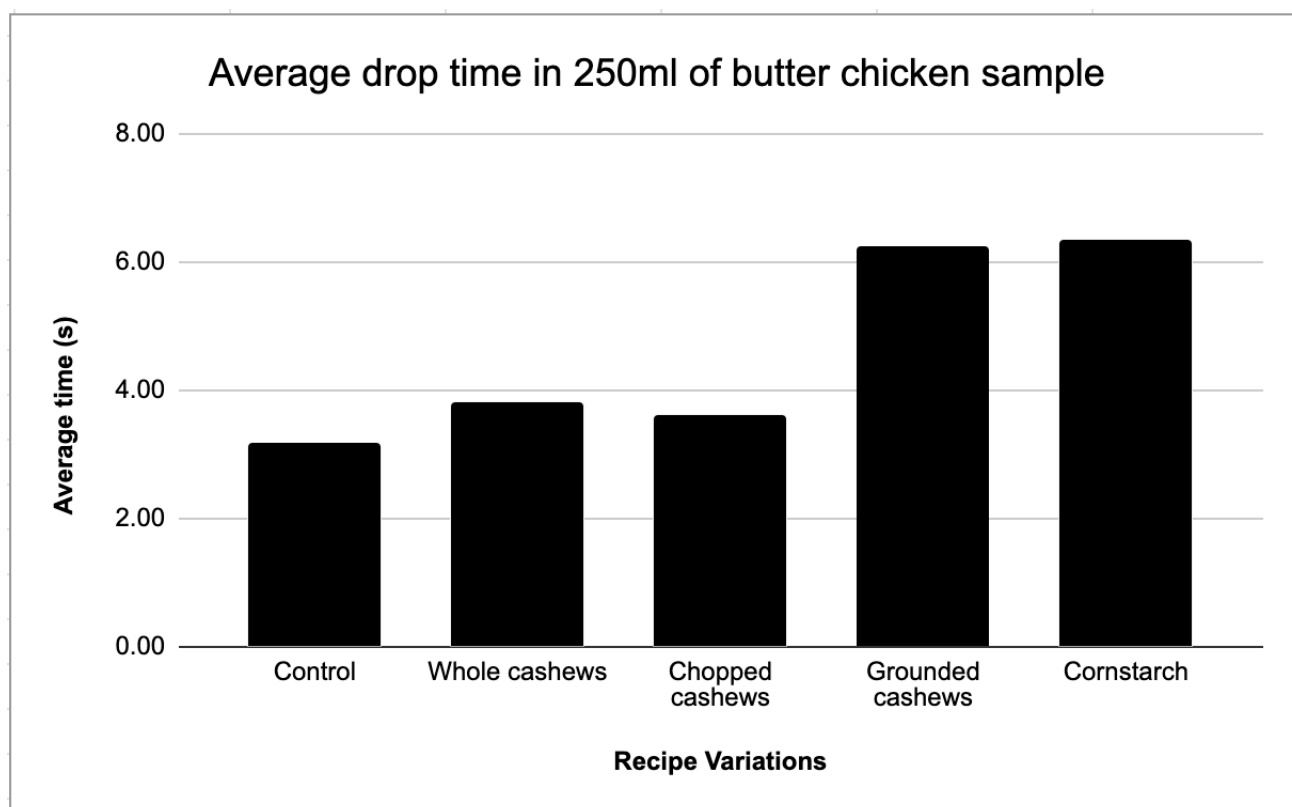
We made 4 variations of butter chicken sauces overall. First, we cooked butter chicken with three types of cashews - whole, chopped or blended with water - and we also made separate sauces without cashews and with cornstarch to compare results. To compare the thickness of the sauces, we measured the time taken for a aluminium cube to drop in 250ml of the sauce. We took three measurements of drop times for each sauce and found an average of the three.

Our discovery

Our findings can be summarised in the bar chart on the next page. We can explain the results with the help of some takeaways from class and our own understanding. Firstly, we learned in class that some substances, when added to a liquid-based dish are able to limit the movement of water molecules as they spread out in a solution - these substances are thus good thickening agents (thickeners). Similarly, when we grind the cashews with water, the hard and crunchy structure falls apart into a paste that contains fats, carbohydrates, and proteins. These molecules are hydrophilic and start to bind with water, preventing them from moving freely and causing the sauce to thicken.

From the bar chart, grounded cashews and cornstarch can be seen to have the greatest thickening effect as they have the highest average drop time. Moreover, in comparison to cornstarch, the gravy made with grounded

cashews was the most flavorful and scored highest in our qualitative taste tests. Therefore, we conclude that the grounded form of cashews is the best thickener because the cashew molecules are able to disperse best in the solution and interrupt the flow of it when grounded.



Bar Chart measuring the thickness upon use of different ingredients.

Conclusion and Lingering Questions

From our findings, we conclude that it is best to use grounded cashews to thicken a butter chicken gravy. This is because grounded cashews not only provide the most effective thickening of the gravy but also taste much more flavourful when compared to cornstarch. Through qualitative feedback, we gathered that using grounded cashews is tastier as it adds a nutty flavour to the gravy compared to the more bland and diluted addition of cornstarch.

As we kept the amount of cashews constant throughout our experiment,

a potential area for further exploration could be to alter the number of cashews in future experiments to determine the ideal number of cashews, in terms of consistency and taste, to use when cooking butter chicken.

Ingredients

Marinade

✳ 1 kg boneless chicken cubes ✳ 4 tbsp chicken tikka masala ✳ 1 tbsp red chilli ✳ 1 tsp salt ✳ 2 tbsp lemon juice ✳ 1 cup yoghurt ✳ 1 tbsp oil

Gravy

✳ 1 cup butter ✳ 1 cup tomato paste ✳ 1/2 cup cashew nuts ✳ 1 cup fresh cream ✳ 1 cup milk ✳ 2 tsp salt ✳ 1 tsp red chili powder ✳ 1 tsp garam masala ✳ 1 tsp dry methi ✳ 1 onion ✳ 4 cloves garlic ✳ 2 TSP grated ginger

Equipment

✳ Induction Cooker ✳ Oven ✳ Pans

Directions

1. Marinate the chicken for 1 hour prior with tikka masala, red chili powder, salt, lemon juice, yogurt, and oil.
2. Grease the tray and cover it with foil. Spread marinated chicken and bake for 20 minutes in preheated oven.
3. Add half a cup of butter to a pan and fry onion and garlic until onions are caramelized.
4. In a separate pan, heat butter and add grounded cashew nut paste,

tomato puree, salt, red chillis, fresh cream, and milk.

5. Let the sauce simmer.
6. When the onion is ready, add the sauce with it, and add garam masala, dry methi, and the cooked chicken.
7. Cover and keep on low heat for 10 minutes; add coriander for garnish.



The finished Butter Chicken, in its creamy and buttery glory!

Hand-pulled Noodles



By Yelani and Hoi Yuet



Prep time 190 minutes, including 60 minutes resting time



Preparation time 20-25 minutes



Serves 2

Summary

Noodles are an essential food in Chinese cuisine, and one of our favorite staples. Furthermore, there are so many different kinds of noodles, with varied elasticity, size, shape and stretchiness (extensibility). In particular, we studied how the resting time of the noodle dough would affect the noodles' stretchiness — how far it can stretch and be pulled until it breaks. We hence varied the resting time that we gave the noodle dough prior to it being pulled. Generally, more resting time correlates to a noodle strand that can stretch further before breaking. These findings suggest that it is not only the hand-pulling technique and elasticity of the noodle that matters in making well-textured noodles, but also how much resting time is given to allow the gluten networks to develop and relax fully.

Why this project

We were wondering how the dough's rest time affects the noodle's stretchability. Every time we go to Haidilao, we are always amazed by how long the staff can pull the noodles. In our past experiences with making noodles, they usually break after a brief attempt to pull them. We wonder

if the resting time of the dough is an important factor. Giving the dough enough time to rest is often a key component in baking and cooking, such as after kneading bread dough. Yet, recipes for noodles rarely mention how long and why we should rest the dough before stretching it out into noodles. Hence, we set off to find out how varying the resting time for the dough can affect its stretchability.

Our experiment

To make the noodle dough, we mixed a consistent amount of bread flour, salt, and water. We then allowed the dough to rest for six different durations: 0 minutes, 15 minutes, 30 minutes, 45 minutes, and 60 minutes. We also left one batch of dough resting overnight. For each batch of dough, we measured the initial length of the noodle strand. We then imitated the Haidilao staff's hand-pulling motions, swinging the thin noodle dough pieces to stretch them evenly. Once the noodle strand was pulled to the point of breaking, we stopped swinging the noodle and measured the final length (including the bit that broke off). We repeated this three times and recorded the average increase in length for each of the setups.

Our discovery

We discovered that resting the dough for 60 minutes yielded the most fantastic average stretch of 40.5cm, almost three times as stretchable as dough with zero resting time. They felt softer to the touch, and we could stretch them easily without exerting much force. In general, our data revealed that the average stretch of the dough increased with resting time. However, there were some inconsistencies in the data collected. For instance, the average stretch for the dough with a 30-minute resting time of 30 minutes was 1.5cm shorter than for the dough with a 15-minute resting time. A possible reason is that we left that 30-minute batch of dough out in the

open, causing it to become dehydrated and, thus, less stretchy.

To further corroborate our findings, we left one dough batch resting overnight. Since we used this batch of dough for our class presentation, we did not stretch this batch of noodles. Observations of the batch of dough overnight include its surface being relatively smooth and the dough being extremely stretchy when we pulled it. Given that the smell of the dough became extremely sour and pungent, we do not recommend leaving the dough overnight to rest.

Gluten consists of glutenin and gliadin. Gliadins are like ball bearings that move freely within the gluten networks and allow glutenin to slide past each other, hence contributing to the stretchability of the dough. During kneading, gluten strands will form, stretch and twist. Long gluten strings are aligned in a stretchy, layered web. The dough becomes stretchier after resting it for longer because it allows the gliadin in the gluten network to relax and settle into the long protein chains. Without rest time, the dough will continue to become very elastic but not stretchy, resulting in a dough that is difficult to shape, roll and pull.

Summary and lingering questions

Just like bread dough in baking, noodle dough also requires resting time. This is because gluten networks which include glutenin and gliadin, require not only kneading and mixing but also resting time to fully develop into an elastic and stretchy dough.

Some questions to explore further include how the cooking oil coating and the amount of salt added to the dough affects the elasticity, stretchability, texture, and taste of the eventual noodle strand, as well as how the type of flour with differing protein contents would relate to the optimum resting time needed.



Two bowls of pulled noodles!

Ingredients

✧ 250g bread flour ✧ 150g water ✧ ¼ teaspoon of salt ✧ Cooking oil
(for coating the dough)

Equipment

✧ Mixing Bowl ✧ Cooking Pot ✧ Rolling Pin ✧ Cling Wrap

Directions

1. Mix bread flour and salt in your mixing bowl. Add in all the water gradually.
2. Knead the mixture briefly until it becomes a dough, then leave it to rest for 15 minutes.
3. Knead again for about 2 minutes, then let it rest for 15 minutes.
4. Repeat step 3.
5. Divide the dough into 2 equal parts.
6. Roll the dough out on a lightly floured surface into round flat sheets, using the rolling pin.
7. Coat the sheets of dough evenly with cooking oil and place them on a tray. Cover them with cling film and let the dough rest for 60 minutes.
8. Cut the dough into strips about 1 cm wide if making round, thin noodles; about 3 cm wide if making flat, wide noodles.
9. Pick up one strip of dough by holding each end gently. Pull towards opposite directions and swing your arms in a forward circular motion consistently until the desired length.

10. Boil water in a big pot, and cook the noodles until desired chewiness and softness! If you prefer softer noodles, cook the noodles for a longer time.
11. Enjoy the noodles with your favorite soup or sauce!



Yummy tomato sauce noodles

The Perfect Eggplant



By Max and Barney



Preparation time 30-35 minutes



Serves 3-4

Summary

Eggplants can be a tasty meal, but too often they are not well cooked in our college dining hall, creating a lot of eggplant food waste. We set out to find a way to ‘perfect’ the texture of an eggplant when preparing it — could changing the preparation method ‘improve’ the eggplant texture? If so, what is the best method? Based on these questions, we tested the changes in elasticity of eggplants when prepared differently prior to cooking. We compared the effects of salt water, a salt rub, and microwaving on the texture of cooked eggplants. Our findings suggest that both methods involving salt were suitable to reach the ‘best’ texture for the eggplant (according to our preferences), yet the taste was negatively affected by the salt rub. We suggest that soaking the eggplant in water with a 3% concentration of salt (14-15g salt per 500ml of water) for at least 10 minutes before cooking is the ideal way of preparing fried eggplants to achieve a texture that is neither too soft nor too rigid.

Why this project

Eggplants are important elements of cuisines all over the world and are a very important element of the everyday diet, particularly for vegetarian

or vegan members of our community. We were concerned about the inconsistent and unsatisfying ways of cooking that we noticed in our college dining hall, and we wanted to prevent these mistakes by testing the best methods to prepare the eggplant prior to cooking. From a culinary perspective, this could be a breakthrough in the way eggplant-based recipes are written. There is no consensus, after all, on how to do it. Some people suggest microwaving, others rub salt, and others soak the eggplant in salt water. For a beginner, it is likely that none of these things are even considered. Scientifically this is an interesting question too since eggplants have an internal sponge-like structure. Given that this structure is not particularly malleable, changing its texture without altering the whole structure is a challenge. Likewise, collapsing the whole structure is not particularly helpful unless aiming to cook some kind of purée (some eggplant dishes like ‘baba ganoush’ or ‘moutabal’ aim for this). Thus, we sought to identify the different processes that take place in the preparation methods we attempted. Although we acknowledge that our results would be arbitrary in some ways (they are based on our preferences), we believe this project would still help us to learn how the eggplant responds to different changes in scientific terms.

Our experiment

To measure the elasticity of the eggplant, we prepared the eggplant in 3 different ways (in addition to having a control set without any kind of preparation) before cooking. We kept the mass (40g) and height (3cm) of our eggplant samples constant, though the shape of the samples had some inevitable variations. We soaked eggplant samples in salt water with three different concentrations (1.5%, 3%, 6%), and we rubbed salt on the eggplant skin in 3 different amounts (2.5g, 5g, 10g); in these cases, we allowed the eggplant sample to rest with the salt/salt water for 10

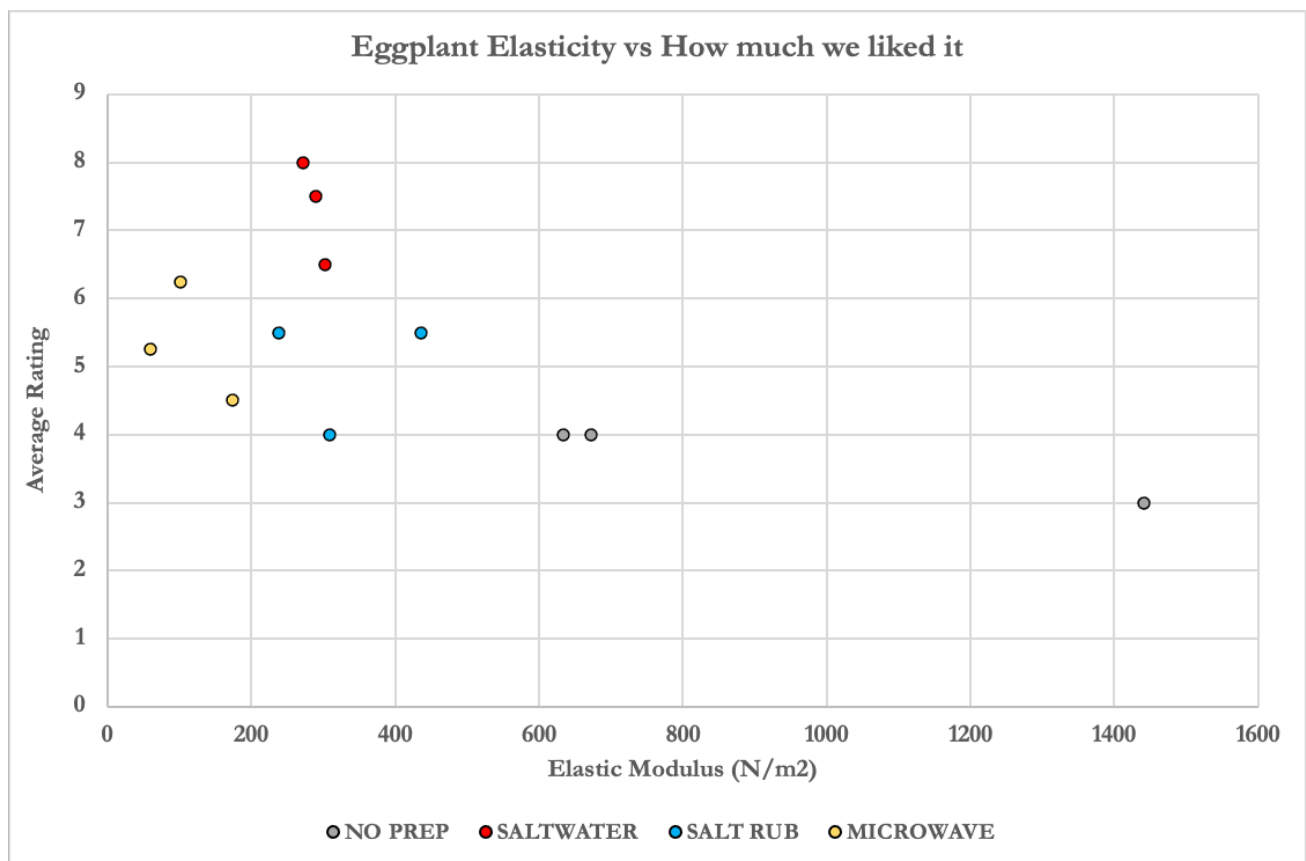
minutes. We also microwaved the eggplant for three different durations (30 seconds, 1 minute, and 90 seconds). We then cooked each sample for 7 minutes at medium heat and used 20ml of olive oil in each case. We allowed the eggplant sample to cool down for 2 minutes and then used a ruler and a weighing scale to record how much force was required to depress each sample by 3mm. We then calculated the elastic modulus of each eggplant sample using Young's modulus.

$$E = \frac{F/A}{\Delta L/L_0}$$

Lastly, we tasted each sample and gave it a 10-point rating based on how much we liked it, noting any relevant observations on its texture and taste.

Our discovery

We discovered that the 'best' way to cook eggplant was soaking it in a saltwater concentration of 3% for 10 minutes (though a longer exposure may give even better results) before cooking it. Compared to the control sample with no prior preparation method, the elastic modulus of the sample that was soaked in saltwater was about 4 times lower, meaning that it had a softer mouthfeel and more appealing texture. The same texture can be achieved through a salt rub, though that method made the eggplant too salty in taste. The microwave also helped soften the eggplant, but we found that it often made it too soft, and we preferred an eggplant that still had a slightly chewy feeling when biting it. In the graph on the next page, you can notice there is almost a bell curve when it comes to the elasticity of the eggplant and how much we liked it, suggesting that there is an ideal zone that is neither too soft nor too hard.



Scatter plot indicating the elastic modulus and average rating of each eggplant sample

The main reason for these differences in elasticity is due to the eggplant's sponge-like structure. Its structure allows it to retain a lot of water and to be very elastic and squishy when biting it. The preparation methods we used all affect the amount of water in the eggplant: both methods involving salt trigger osmosis in the eggplant, so it releases the liquid it has absorbed. We were initially surprised when we noticed some unexpected water coming out from the eggplant after rubbing it in salt. Likewise, the microwave pre-cooks it and makes some of the water evaporate. The levels of water in the eggplant have effects on its texture primarily because they change what happens when we bite the eggplant. When there is water, we need more force to bite the eggplant because we first need to squish the water out before we can tear the structure, which makes it chewy and somewhat rigid/hard. When we remove some moisture, we squish only air pockets (or at least less liquid), making it easier to tear the structure.

However, it is also important to keep the structure somewhat stable, or else the eggplant might feel too creamy and not chewy enough. Ultimately, it depends on the aims of the recipe.

Conclusion and lingering questions

The ‘perfect’ eggplant is a subjective ideal. It is based on each individual’s preference and the intent behind the cooking of the eggplant. However, we found that different methods can help us achieve the intended textures with more ease than just trying to cook the eggplant for longer times. If we want a soft eggplant, microwaving it before we fry it makes it easy! If we want a nice, soft, and somewhat chewy eggplant, then exposing it to salt and letting it rest for some time before cooking also works well. These are primarily ways of making the cooking process easier, faster, and more predictable, allowing us to make informed decisions about the final texture we want to achieve. It remains unclear whether the time of exposure to salt for the eggplant significantly affects its texture. Likewise, we do not know enough about the eggplant anatomy to tell if cooking different parts—or applying salt in different parts—matters or not. We have learned that cooking an eggplant without prior preparation may make it difficult to achieve the ideal texture. Hopefully, our experimentation and data can help someone cook their perfect eggplant as well.

Ingredients

✳️ 1 Eggplant ✳️ 75ml olive oil ✳️ 500g water ✳️ Half a Lemon ✳️ 14g (about 2tsp) salt ✳️ 4 cloves of garlic ✳️ 1 Carrot ✳️ Truffle and Tomato sauce *We recommend plating the eggplant with mushrooms and olives



The perfect eggplant slices, plated with mushrooms, olives, and baba ganoush!

Equipment

✧ Induction Cooker ✧ Frying Pan

Directions

1. Rinse your lemon, eggplant, and carrot.
2. Slice the eggplant into long yet somewhat thin slices. Aim for 1.5cm or about the thickness of a burger patty. Soak the eggplant slices in a salt-water solution. In 500ml of water, there should be about 2 teaspoons of salt. Let it soak for 10-15 minutes.
3. Peel and chop your garlic and carrots into very small cubes.

4. (This step should be done at most 7 minutes before the eggplants have completed their soaking period to avoid overcooking the carrot and garlic) Turn on your stove and add olive oil. Let it heat and once hot add your garlic. When the garlic starts to acquire a light brown color, add the carrots.
5. Add the eggplant slices and flip them around approximately every 45 seconds. Squeeze some of the lemon halves periodically until it has all been squeezed. Cook the eggplant until the skin looks soft and there is browning all over. All of the skin should acquire at least a dark green look, and the skin should look a little bit wrinkly in all places before it is considered done.
6. Plate the eggplant and add a thin layer of tomato and truffle sauce below it to give it an aromatic feel.



Special thanks to:

*The Yale-NUS Science Division and the Physical Sciences Major for making
this course possible*

Dr. Leow Yu Hui for supporting all the experiments in this course

*Saga Buttery Team and the Saga College Office for providing the space and
the support*

Yale-NUS College Infrastructure, Safety & Security for ensuring our safety

*Chef Dean Brettschneider, Chef Jenna White, and the team at Brettschneider's
Baking & Cooking School, for teaching us so much about the art and science
of bread baking*

*The Teaching Engagement Grant at Yale-NUS, for providing the financial
support for the bread-baking workshop*

*The Faculty Support Office at Yale-NUS, for providing the logistic support for
the course*





EXPERIMENTS

- 1: Ricotta cheese
- 2: Tempering chocolate
- 3: Ice cream in Iced water
- 4: Mayonnaise and Egg-White Foam
- 5: Molten Chocolate cake
- 6: Milkshakes
- 7: Vanilla Flan
- 8: Bread

$$pH = -\log[H^+]$$

$$N(t) = N_0 e^{kt}$$

$$E = \frac{k_B T}{f^3}$$

$$L = \sqrt{4Dt}$$

$$Q = mc\Delta T$$